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Refrigerator and elevation device for refrigerator

Abstract

An elevation device for a refrigerator drawer includes a lower frame, an upper frame, and a lifting assembly. The lifting assembly includes: a pair of first rods having a first end rotatably coupled to the lower frame and a second end that translates along the upper frame; and a pair of second rods having a first end rotatably coupled to the upper frame and a second end that translates along the lower frame. The second rod is rotatably coupled to and crosses the first rod. The first or second rod is connected to a driving device via the first end of the first or second rod, with the driving device disposed outside the upper frame and the lower frame. The first and second rods are configured to, based on power from the driving device, rotate about their respective first ends to elevate the upper frame relative to the lower frame.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. application Ser. No. 18/140,238, filed on Apr. 27, 2023, which is a continuation of U.S. application Ser. No. 17/673,526, filed on Feb. 16, 2022, now U.S. Pat. No. 11,668,517, which is a continuation of U.S. application Ser. No. 17/126,468, filed on Dec. 18, 2020, now U.S. Pat. No. 11,287,178, which is a continuation of U.S. application Ser. No. 16/230,474, filed on Dec. 21, 2018, now U.S. Pat. No. 10,921,052, which claims priority under 35 U.S.C. 119 and 35 U.S.C. 365 to Korean Patent Application No. 10-2018-0071898, filed on Jun. 22, 2018. The disclosures of the prior applications are incorporated by reference in their entirety.

BACKGROUND

(1) The present invention relates to a refrigerator and an elevation for a refrigerator.
(2) In general, refrigerators are home appliances for storing foods at a low temperature in a storage space that is covered by a door. For this, refrigerators cool the inside of the storage space by using cool air generated by being heat-exchanged with a refrigerant circulated through a refrigeration cycle to store foods in an optimum state.

- (3) In recent years, refrigerators have become increasingly multi-functional with changes of dietary lives and gentrification of products, and refrigerators having various structures and convenience devices for convenience of users and for efficient use of internal spaces have been released.
- (4) The storage space of the refrigerator may be opened/closed by the door. Also, refrigerators may be classified into various types according to an arranged configuration of the storage space and a structure of the door for opening and closing the storage space.
- (5) The refrigerator door may be classified into a rotation-type door that opens and closes a storage space through rotation thereof and a drawer-type door that is inserted and withdrawn in a drawer type.
- (6) Also, the drawer-type door is often disposed in a lower region of the refrigerator. Thus, when the drawer-type door is disposed in the lower region of the refrigerator, a user has to turn its back to take out a basket or foods in the drawer-type door. If the basket or the foods are heavy, the user may feel inconvenient to use the basket or may be injured.
- (7) In order to solve such a limitation, various structures are being developed in which the drawer-type door is capable of being elevated.
- (8) Representatively, a refrigerator in which a lifting mechanism for elevation a bin provided in a refrigerating compartment is disclosed in U.S. Pat. No. 9,377,238.
- (9) However, this technique according to related art may have a structure in which the lifting mechanism for the elevation is disposed and exposed outside the bin to cause a serious safety problem. Also, there is a limitation that an outer appearance is poor due to the structure of the lifting mechanism exposed to the outside.
- (10) Also, since a driving part is exposed to the outside, noise during operation of the driving part may be transmitted to the outside as it is, which may cause the user's dissatisfaction.
- (11) Also, since the lifting mechanism is disposed inside the refrigerator, storage capacity within the refrigerator may be significantly reduced. This may cause a limitation that storage efficiency of the refrigerator is greatly reduced due to the loss in storage capacity of the whole refrigerator.
- (12) Also, the lifting mechanism is provided inside the refrigerator. Thus, separation of the door and separation of the lifting mechanism are required for service of the lifting mechanism to deteriorate serviceability.
- (13) Also, a driving part of the lifting mechanism has a structure for elevating the bin by pushing one end of the support assembly. Therefore, when a large heavy structure or a heavy object is disposed inside the bin, sufficient force for the elevation may not be provided. Of course, although a motor of the driving part increases in size to solve this limitation, there are limitations that an internal volume loss and noise become larger, and the manufacturing cost increases.
- (14) Also, the lifting mechanism supports one side of the entire bottom surface of the bin due to the arrangement position of the driving part. Therefore, in the state where the bin is filled with a stored product, an eccentric load may occur. Here, a serious problem may arise in stability due to the eccentric load acting in a state in which the door is withdrawn, there is a limitation that the elevation operation is not performed smoothly.
- (15) Also, the lifting mechanism has a structure in which the entire bin is elevated. In order to elevate the bin, the bin has to be completely withdrawn from the storage space of the refrigerator. Also, when the bin is elevated, the bin has to be withdrawn up to a position at which the bin does not interfere with the upper door and the refrigerator body. However, in this structure, when the door is completely withdrawn, a loss of cold air within the refrigerator may cause a limitation in stability, and there is a possibility that stability is deteriorated by an occurrence of deflection due to the load of the lifting mechanism. Thus, it is necessary to supplement the draw-out structure, and there is a limitation in that it is difficult to be applied to the structure of the bin or door which is substantially large in size.

SUMMARY

- (16) Embodiments provide a refrigerator in which an electric device for elevation is provided inside

a door part, and a mechanical device for the elevating the drawer part is provided in a drawer outside the door and an elevation for a refrigerator.

(17) Embodiments also provide a refrigerator which is capable of preventing deflection from occurring by an eccentric load when the drawer part is elevated to ensure a stable elevation operation and an elevation for a refrigerator.

(18) Embodiments also provide a refrigerator in which an elevation device having a mechanism structure, which is provided in a drawer part to elevate at least a portion of the drawer part, and a driving device that is an electrical device, which is provided in a door part to generate power, are separated together with each other when the door part and the drawer part are separated and an elevation device for a refrigerator.

(19) Embodiments also provide a refrigerator which is improved in assembly workability, cleanability, and serviceability of a drawer door that is capable of being inserted and withdrawn and an elevation for a refrigerator.

(20) Embodiments also provide a refrigerator which is capable of providing a withdrawable structure in a state in which a loss of storage capacity is minimized and an elevation for a refrigerator.

(21) Embodiments also provide a refrigerator, which is smoothly elevated by a lifting assembly and an elevation device for a refrigerator.

(22) Embodiments also provide a refrigerator which has a structure in which a lifting assembly is not exposed to the outside when an elevation device is elevated to improve an outer appearance and safety in use and an elevation device for a refrigerator.

(23) According to one aspect of the subject matter described in this application, an elevation device for a refrigerator includes: a lower frame configured to be fixed to a bottom of a drawer, the drawer being configured to insert into and withdraw from the refrigerator; an upper frame disposed vertically above the lower frame and configured to support an object stored in the drawer; and a lifting assembly. The lifting assembly includes: a pair of first rods including a first rod, the pair of first rods having a first end rotatably coupled to the lower frame and a second end configured to translate along the upper frame; and a pair of second rods including a second rod, the pair of second rods having a first end rotatably coupled to the upper frame and a second end configured to translate along the lower frame. The second rod is rotatably coupled to the first rod, and crosses the first rod. The first rod or the second rod is connected to a driving device via the first end of the first rod or the second rod, with the driving device disposed outside the upper frame and outside the lower frame and configured to transmit power. The pair of first rods and the pair of second rods are configured to, based on the power transmitted from the driving device, rotate about their respective first ends to elevate the upper frame relative to the lower frame.

(24) Implementations according to this aspect may include one or more of the following features. For example, the lifting assembly is a first lifting assembly configured to support a first side of the upper frame, and the elevation device further includes a second lifting assembly configured to support a second side of the upper frame. The second lifting assembly may include a pair of third rods and a pair of fourth rods, and the first lifting assembly and the second lifting assembly are connected to the driving device and configured to be simultaneously operated by the driving device.

(25) In some implementations, the upper frame includes an upper frame edge that extends from a circumference of the upper frame toward the lower frame, and the lower frame includes a lower frame edge that extends from a circumference of the lower frame toward the upper frame. The upper frame may be configured to: move downward to the lower frame; based on moving downward to a lowest position, contact the lower frame; and based on contacting the lower frame, define an accommodation space configured to accommodate the pair of first rods and the pair of second rods.

(26) In some implementations, one of the upper frame edge or the lower frame edge includes a

coupling protrusion, where the other of the upper frame edge or the lower frame edge defines a coupling groove configured to receive the coupling protrusion, the coupling groove having a shape corresponding to the coupling protrusion. The coupling groove and the coupling protrusion are configured to couple to each other in a state in which the upper frame contacts the lower frame. In some examples, the pair of first rods and the pair of second rods are configured to overlap each other, where, in a state in which the pair of first rods and the pair of second rods overlap each other, a height of the accommodation space is greater than a sum of a thickness of the pair of first rods and a thickness of the pair of second rods.

(27) In some implementations, each of the upper frame and the lower frame has a rectangular frame shape with an opened central portion, where the elevation device further includes a support plate that is located on the upper frame, that covers the upper frame, and that is configured to support the object stored in the drawer. In some examples, the support plate includes: an edge part located at a perimeter of the support plate and configured to accommodate the upper frame; and a support part that is surrounded by the edge part, that is recessed toward the lower frame to store a food item or a container therein, and that is configured to insert into the opened central portion of each of the upper frame and the lower frame.

(28) In some implementations, the elevation device further includes a rotation shaft that passes through the lower frame, that protrudes from the lower frame to an outside of the lower frame, and that is configured to be connected to the driving device at the outside of the lower frame, where the first rod is configured to rotate about the rotation shaft at the first end of the first rod. In some examples, the first rod includes a rod protrusion that protrudes from a side of the first rod and that is spaced apart from the rotation shaft, and the first rod is configured to rotate about the rotation shaft in a state in which both of the rotation shaft and the rod protrusion are coupled to the driving device.

(29) In some implementations, the elevation device further includes: a first sliding shaft that connects the second ends of the pair of first rods to each other, the pair of first rods being spaced apart from each other; and a second sliding shaft that connects the second ends of the pair of second rods to each other. The pair of second rods may be spaced apart from each other. The upper frame may define a first slide guide at an inner surface of the upper frame in which the first slide guide is configured to receive the first sliding shaft and to guide a movement of the first sliding shaft. The lower frame may define a second slide guide at an inner surface of the lower frame in which the second slide guide is configured to receive the second sliding shaft and to guide a movement of the second sliding shaft.

(30) In some examples, the elevation device further includes an elastic member that connects the first sliding shaft to a side of the upper frame facing the first sliding shaft, where the elastic member is configured to be tensioned based on the upper frame moving toward the lower frame. In some examples, each of the upper frame and the lower frame defines an opening, where the elevation device further includes: an upper partition part that crosses the opening of the upper frame; and a lower partition part that crosses the opening of the lower frame. The upper partition part and the lower partition part may be configured to face each other and to divide each of the opening of the upper frame and the opening of the lower frame into a left opening and a right opening. The pair of first rods and the pair of second rods may be disposed at the left opening, and the elevation device may further include a pair of third rods and a pair of fourth rods disposed at the right opening.

(31) In some implementations, the elevation device may further include a roller disposed at the second end of each of the first rod and the second rod, where the roller at the second end of the first rod is configured to contact and roll along the upper frame based on rotation of the first rod about the first end of the first rod. The roller at the second end of the second rod may be configured to contact and roll along the lower frame based on rotation of the second rod about the first end of the second rod.

(32) According to another aspect, a refrigerator includes: a cabinet that defines an upper storage space and a lower storage space; a front panel door part configured to open and close the lower storage space; a drawer part configured to insert into and withdraw from the lower storage space; a driving device disposed at the front panel door part; and an elevation device disposed at the drawer part and configured to elevate an object stored in the drawer part. The elevation device includes: a lower frame disposed inside the drawer part; an upper frame disposed vertically above the lower frame and configured to support the object; and a lifting assembly including a plurality of rods that are rotatably coupled to each other, that cross each other, and that connect the lower frame to the upper frame. The driving device is configured to drive a rotation of at least one rod among the plurality of rods, through an end of the at least one rod that is rotatably coupled to the lower frame or to the upper frame, to elevate the upper frame relative to the lower frame.

(33) Implementations according to this aspect may include one or more of the following features or the features described above. For example, the drawer part may define a drawer space having an opening at a top of the drawer space, where the drawer space includes: a front space at which the elevation device is arranged, the front space being configured to be positioned at an outside of the lower storage space based on the drawer part being withdrawn from the lower storage space; and a rear space defined rearward of the front space. Each of the upper frame and the lower frame may have a size corresponding to a size of the front space.

(34) In some implementations, the refrigerator further includes a drawer cover located in the drawer part and configured to partition the drawer space into the front space and the rear space. In some implementations, the elevation device further includes a connection assembly that is located at the front panel door part, that is configured to couple to the elevation device exposed at a front surface of the drawer part, and that is configured to transmit power from the driving device to the elevation device. The connection assembly may be configured to selectably separate the driving device at the front panel door part from the elevation device at the drawer part.

(35) In some implementations, each of the upper frame and the lower frame has a rectangular frame shape with an opened central portion, where the upper frame is configured to: move downward to the lower frame; based on moving downward to the lower frame, contact the lower frame; and based on contacting the lower frame, define an accommodation space configured to accommodate the plurality of rods.

(36) According to another aspect, a refrigerator includes: a cabinet that defines an upper storage space and a lower storage space; a door assembly that is configured to open and close the lower storage space and that is configured to insert into and withdraw from the lower storage space; a driving device disposed at a portion of the door assembly; an elevation device support part that extends from a rear surface of the door assembly toward the lower storage space; an elevation device disposed on the elevation device support part and configured to be elevated by the driving device; and a basket disposed on the elevation device and configured to be elevated by the elevation device. The elevation device includes: a lower frame fixed to the elevation device support part; an upper frame disposed vertically above the lower frame and configured to support the basket; and a lifting assembly including a plurality of rods that are rotatably coupled to each other, that cross each other, and that connect the lower frame to the upper frame. The driving device is configured to drive a rotation of at least one rod among the plurality of rods, through an end of the at least one rod that is rotatably coupled to the lower frame or to the upper frame, to elevate the upper frame relative to the lower frame.

(37) Implementations according to this aspect may include one or more of the features described above or the following features. For example, the elevation device support part may include a surface configured to support a bottom surface of the elevation device.

(38) The details of one or more embodiments are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a front view of a refrigerator according to a first embodiment.
- (2) FIG. 2 is a schematic view illustrating a state in which a drawer door of the refrigerator is elevated.
- (3) FIG. 3 is a perspective view illustrating a state in which a container of the drawer door is separated.
- (4) FIG. 4 is an exploded perspective view illustrating a state in which the drawer part of the drawer door and the front panel door part are separated from each other when viewed from a front side.
- (5) FIG. 5 is a rear view of the front panel door part.
- (6) FIG. 6 is a rear view illustrating a state in which a door cover of the front panel door part is removed.
- (7) FIG. 7 is an exploded perspective view of the front panel door part.
- (8) FIG. 8 is a perspective view of a door device according to a first embodiment.
- (9) FIG. 9 is an exploded perspective view of the driving device.
- (10) FIG. 10 is a cross-sectional view of the screw assembly that is one component of the driving device.
- (11) FIG. 11 is an exploded perspective view of the screw assembly.
- (12) FIG. 12 is an exploded perspective view of the motor assembly that is one component of the driving part.
- (13) FIG. 13 is a view illustrating a coupling structure of the motor assembly and a driving shaft.
- (14) FIG. 14 is an exploded perspective illustrating a coupling structure of a connection assembly, which is one component of the driving device, and a lever.
- (15) FIG. 15 is an exploded perspective view of the connection assembly when viewed in one direction.
- (16) FIG. 16 is an exploded perspective view of the connection assembly when viewed in the other direction.
- (17) FIGS. 17 and 18 are views illustrating an operation state of the connection assembly.
- (18) FIG. 19 is an exploded perspective view of the drawer part.
- (19) FIG. 20 is an exploded perspective view illustrating a coupling relationship between the drawer part and the connection assembly.
- (20) FIG. 21 is an enlarged view illustrating a portion A of FIG. 20.
- (21) FIG. 22 is a front view of an elevation device according to the first embodiment.
- (22) FIG. 23 is an exploded perspective view illustrating a state in which a support plate is separated from the elevation device.
- (23) FIG. 24 is a perspective view of the elevation device.
- (24) FIG. 25 is an exploded perspective view of the elevation device in a state in which a lifting assembly that is one component of the elevation device is unfolded.
- (25) FIG. 26 is an exploded perspective view of the elevation device in a state in which the lifting assembly is folded.
- (26) FIG. 27 is a perspective view of an upper frame that is one component of the elevation device.
- (27) FIG. 28 is a perspective view of the lifting assembly.
- (28) FIG. 29 is a perspective view illustrating a state in which the elevation device ascends when viewed from a lower side.
- (29) FIG. 30 is an enlarged view of a portion “B” of FIG. 29.
- (30) FIG. 31 is an enlarged view of a portion “C” of FIG. 29.
- (31) FIG. 32 is a partial perspective view of one side of a lower portion in the state in which the

elevation device ascends.

(32) FIG. **33** is a cross-sectional view taken along line **33-33'** of FIG. **23**.

(33) FIG. **34** is a perspective view illustrating a connection state between the connection assembly and the elevation device.

(34) FIG. **35** is a cross-sectional view illustrating the connection state between the connection assembly and the elevation device.

(35) FIG. **36** is a perspective view illustrating a separation state of the connection assembly and the elevation device.

(36) FIG. **37** is a perspective view illustrating a state in which the drawer door is closed.

(37) FIG. **38** is a perspective view illustrating a state in which the drawer door is completely opened.

(38) FIG. **39** is a cross-sectional view illustrating a state of the drawer door in a state in which the basket of the drawer door completely descends.

(39) FIG. **40** is a perspective view illustrating a state of the driving device in the state in which the basket of the drawer door completely descends.

(40) FIG. **41** is a perspective view illustrating a state of the elevation device in the state in which the basket of the drawer door completely descends.

(41) FIG. **42** is a cross-sectional view illustrating a state of the drawer door in a state in which the basket of the drawer door completely ascends.

(42) FIG. **43** is a perspective view illustrating a state of the driving device in the state in which the basket of the drawer door completely ascends.

(43) FIG. **44** is a perspective view illustrating a state of the elevation device in the state in which the basket of the drawer door completely ascends.

(44) FIG. **45** is an exploded perspective view illustrating a coupling structure of an elevation device and a support plate according to a second embodiment.

(45) FIG. **46** is an exploded perspective view of the elevation device.

(46) FIG. **47** is a view illustrating an arrangement of a support plate in a state in which the elevation device descends at the lowest position.

(47) FIG. **48** is a view illustrating an operation of an elevation device of a refrigerator according to a third embodiment.

(48) FIG. **49** is a perspective view of a refrigerator according to a fourth embodiment.

(49) FIG. **50** is a perspective view of a refrigerator according to a fifth embodiment.

(50) FIG. **51** is a perspective view of a refrigerator according to a sixth embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(51) Hereinafter, detailed embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. However, the scope of the present disclosure is not limited to proposed embodiments, and other regressive inventions or other embodiments included in the scope of the spirits of the present disclosure may be easily proposed through addition, change, deletion, and the like of other elements.

(52) FIG. **1** is a front view of a refrigerator according to a first embodiment. Also, FIG. **2** is a schematic view illustrating a state in which a drawer door of the refrigerator is elevated.

(53) As illustrated in the drawing, the refrigerator **1** may have an outer appearance that is defined by a cabinet **10** defining a storage space and a door **2** covering an opened front surface of the cabinet **10**.

(54) The storage space of the cabinet **10** may be divided into a plurality of spaces. For example, an upper space **11** of the cabinet **10** may be provided as a refrigerating compartment, and a lower space **12** may be provided as a freezing compartment. Each of the upper space and the lower space may be provided as an independent space that is maintained at a different temperature, except for the refrigerating compartment and the freezing compartment. The upper space and the lower space may be called an upper space and a lower space.

(55) The door **2** may be constituted by a rotation door **20** opening and closing the upper space through rotation thereof and a drawer door **30** opening and closing the lower space by being inserted or withdrawn in a drawer type. The lower space may be vertically divided again. The drawer door **30** may be constituted by an upper drawer door **30** and a lower drawer door **30**. Also, an outer appearance of each of the rotation door **20** and the drawer door **30** may be made of a metal material and be exposed to the front side.

(56) Although the refrigerator in which all of the rotation door **20** and the drawer door **30** are provided is described, the present disclosure is not limited thereto. For example, the present disclosure may be applied to all refrigerators including a door that is inserted and withdrawn in the drawer type. Also, the rotation door **20** may be provided at an upper portion and thus called an upper door, and the drawer door **30** may be provided at a lower portion and thus called a lower door.

(57) A display **21** may be disposed on one side of a front surface of the rotation door **20**. The display **21** may have a liquid crystal display structure or a **88** segment structure. Also, when the outer appearance of the door **2** is made of the metal material, a plurality of fine holes are punched in the display **21** to display information by using light passing therethrough.

(58) Also, a manipulation part **22** that is capable of manipulating automatic rotation or withdrawal of the upper door **2** or the lower door **2** may be provided on one side of the rotation door **20**. The manipulation part **22** may be integrated with the display **21** and may operate in a touch manner or a button manner. The manipulation part **22** may input an overall operation of the refrigerator **1** and manipulate an insertion and withdrawal of the drawer door **30** or an elevation within the drawer door.

(59) A manipulation part **301** may also be provided on the drawer door **30**. The manipulation part **301** may be disposed on one side of the drawer door **30** that is disposed at the lowermost portion of the drawer door **30**. The manipulation part **301** may operate in a touch or button manner.

(60) The manipulation part **22,301** may be provided as a sensor detecting proximity or movement of a user or provided as an input unit that operates by a user's motion or voice.

(61) Also, as illustrated in the drawings, an inclined part **311a** may be disposed on a lower portion of a front surface of the lower drawer door **30**, and a manipulation device **302** may be mounted on the inclined part **311a**. The manipulation device **302** may include a projector light capable of outputting an image and a proximity sensor and may project a virtual switch on the floor in the form of an image to detect the image by the proximity sensor. Of course, the manipulation device **302** may be constituted simply by only a proximity sensor. An automatic insertion and withdrawal and/or elevation of the lower drawer door **30** may be manipulated by the manipulation device.

(62) The lower drawer door **30** may be automatically inserted and withdrawn according to the manipulation of the manipulation part **301**. Also, a food or container **36** within the lower drawer door **30** may be elevated in a state in which the drawer door **30** is withdrawn by the manipulation of the manipulation part **301**.

(63) That is, the automatic insertion and withdrawal and/or automatic elevation of the lower drawer door **30** may be performed by at least one of a plurality of manipulation devices **22, 301, 302**, and **303**. As necessary, only one of the plurality of manipulation devices **22, 301, 302**, and **303** may be provided.

(64) Also, a manipulation device (see reference numeral **303** of FIG. **3**) may be provided on a top surface of the lower drawer door **30**. When the manipulation device **303** is provided on the upper surface of the lower drawer door **30**, the lower drawer door **30** may not be manipulated because the lower drawer door **30** is not exposed in the closed state. Thus, the manipulation device **303** may be used for elevating the lower drawer door **30**.

(65) The manipulation devices **22, 301, 302**, and **303** may be used to insert/withdraw and elevate the lower drawer door **30**. Also, the insertion/withdrawal and the elevation may be performed by a combination or sequential operation of the plurality of manipulation devices **22, 301, 302**, and **303**.

(66) The lower drawer door **30** may be a storage space defined in a lower side of the refrigerator **1** and may withdraw the lower drawer door **30** forward to accommodate a food stored in the lower drawer door **30**, and then, the container **36** inside the drawer door **30** may be manipulated to be elevated.

(67) The container **36** may have a predetermined height. Since the container **36** is seated on the elevation device **80**, the height of the container **36** may increase by the height of the elevation device **80** when the elevation device **80** is elevated. Thus, when the elevation device **80** ascends, the user may easily accessible to the container **36** and very easily lift the container **36**.

(68) The container **326** may be completely accommodated in the accommodation part **32** when the door **30** is inserted and withdrawn. When the elevation device **80** ascends, the elevation device may be disposed at a position that is higher than that of an upper end of the drawer door **30**.

(69) The lower drawer door **30** may be automatically inserted and withdrawn forward and backward by the draw-out motor **14**, the pinion **141** provided in the cabinet **10**, and the draw-out rack **34** provided on the bottom surface of the lower drawer door **30**.

(70) Also, the container inside the lower drawer door **30** may be elevated by the driving device **40** and the elevation device **80** provided in the lower drawer door **30**.

(71) Hereinafter, the lower drawer door **30** and an operation of the lower drawer door **30** will be described in more detail, and also, the lower drawer door **30** will be called a drawer door or a door unless otherwise specified.

(72) The embodiments are not limited to the number and shape of the drawer doors **30** and may be applied to all refrigerators having a door that is inserted and withdrawn in a drawer type into/from the lower storage space.

(73) FIG. **3** is a perspective view illustrating a state in which a container of the drawer door is separated. FIG. **4** is an exploded perspective view illustrating a state in which the drawer part of the drawer door and the front panel door part are separated from each other when viewed from a front side.

(74) As illustrated in the drawings, the door **30** may include a front panel door part **31** opening and closing the storage space and a drawer part **32** coupled to a rear surface of the front panel door part **31** and inserted and withdrawn together with the front panel door part **31**.

(75) The front panel door part **31** may be exposed to the outside of the cabinet **10** to define an outer appearance of the refrigerator **1**, and the drawer part **32** may be disposed inside the cabinet **10** to define an storage space. Also, the front panel door part **31** and the drawer part **32** may be coupled to each other and inserted and withdrawn forward and backward together with each other.

(76) The drawer part **32** may be disposed on the rear surface of the front panel door part **31** to define a space in which the food or container to be stored is accommodated. The inside of the drawer part **32** may provide an upwardly opened storage space, and an outer appearance of the drawer part **32** may be defined by a plurality of plates (see reference numerals **391**, **392**, and **395** in FIG. **19**). Each of the plurality of plates **391**, **392**, and **395** may be made of a metal material and provided inside and outside the drawer part **32** so that the entire drawer part **32** is made of stainless steel or a material having a texture such as stainless steel.

(77) In the state in which the door **30** is inserted, a machine room **3** in which a compressor and a condenser constituting a refrigeration cycle are provided may be disposed behind the door **30**. Thus, a rear end of the drawer part **32** may have a shape of which an upper end further protrudes from a lower end, and an inclined surface **321** may be provided on a rear surface of the drawer part **32**.

(78) Also, a draw-out rail **33** guiding the insertion and withdrawal of the door **30** may be provided on each of both side surfaces of the drawer part **32**. The door **30** may be mounted to be inserted into or withdrawn from the cabinet **10** by the draw-out rail **33**. The draw-out rail **33** may be covered by an outer side plate **391** and thus may not be exposed to the outside. The draw-out rail **33** may have a rail structure that is capable of extending in multistage.

(79) A rail bracket **331** may be provided in the draw-out rail **33**, and the rail bracket **331** may extend from one side of the draw-out rail **33** to both sides of the drawer part **32**. Also, the rail bracket **331** may be fixedly coupled to a sidewall surface inside the refrigerator. Thus, the drawer part **32**, that is, the door **30**, may be mounted to the cabinet **10** by the draw-out rails **33**.

(80) Also, the draw-out rail **33** may be provided on a lower end of each of both the side surfaces of the drawer part **32**. Thus, it may be understood that the draw-out rail **33** is disposed on the bottom surface of the drawer part **32**. Thus, the draw-out rail **33** may be provided at a lower ends of each of both sides of the drawer part **32** and may be called an under rail.

(81) A draw-out rack **34** may be disposed on the bottom surface of the drawer part **32**. The draw-out rack **34** may be disposed on each of both sides and be interlocked with an operation of a draw-out motor **14** mounted on the cabinet **10** to automatically insert and withdraw the door **30**. That is, when an operation is inputted into the manipulation parts **22** and **301**, the draw-out motor **14** may be driven to insert and withdraw the door **30** according to movement of the draw-out rack **34**. Here, the door **30** may be stably inserted and withdrawn by the draw-out rail **33**.

(82) The draw-out rack **34** may not be provided on the drawer part **32**. Here, the user may hold a side of the front panel door part **31** to push and pull the front panel door part **31** so that the door **30** is directly inserted and withdrawn.

(83) The inside of the drawer part **32** may be divided into a front space **S1** and a rear space **S2**. The elevation device **80** that is vertically elevated and a container seated on the elevation device **80** to be elevated together with the elevation device **80** may be disposed in the front space **S1**. Although the container **36** is illustrated in the form of a basket having an opened upper portion, the container **36** may have a closed box structure such as a kimchi box. Also, a plurality of containers **36** may be stacked or arranged in parallel to each other.

(84) Also, when the door **30** is withdrawn, the entire drawer part **32** may not be withdrawn to the outside of the storage space due to a limitation in draw-out distance of the door **30**. That is, at least the front space **S1** is withdrawn to the outside of the storage space, and the whole or a portion of the rear space **S2** is disposed inside the storage space within the cabinet **10**.

(85) As described above, a distance of the door **30** may be limited by the draw-out rack **34** or the draw-out rail **33**. As the draw-out distance of the door **30** increases, the door **30** may have large moment applied to the door **30** in a draw-out state, which makes it difficult to maintain a stable state, and the draw-out rail **33** or the draw-out rack **34** may be deformed or damaged, or the refrigerator may be fell or unstable.

(86) The elevation device **80** and the container **36** may be accommodated in the front space **S1**. While the elevation device is elevated, the food or container **36** seated on the elevation device **80** may be elevated together. Also, the elevation device **80** may be provided below the container **36**, and the elevation device **80** may be covered by the container **36** when the container **36** is mounted. Thus, any constituent of the elevation device **80** will not be exposed to the outside.

(87) A separate drawer cover **37** may be provided in the rear space **S2**. The front space **S1** and the rear space **S2** may be partitioned by the drawer cover **37**. In a state in which the drawer cover **37** is mounted, a space in which front and top surfaces of the rear space **S2** are covered and not be used may be not be exposed to the outside.

(88) However, when the drawer cover **37** is separated, the user may be accessible to the rear space **S2**, and thus, foods may be easily accommodated in the rear space **S2**. To utilize the rear space **S2**, a separate pocket or a container corresponding to the shape of the rear space may be disposed in the rear space **S2**.

(89) Also, the elevation device **80** inside the drawer part **32** may be simply separated and mounted to utilize the entire space inside the drawer part **32**, and the elevation device **80** and the drawer cover **37** may be separated from each other to utilize the entire space of the drawer part **32**.

(90) Here, the elevation device **80** may be separated from the driving device **40**. Thus, the elevation device may be simply separated to the side of the drawer part **32** without separating the driving

device **40**. Separation and mounting structures of the elevation device **80** will be described below in more detail.

(91) The outer appearance of each of the inner and outer surfaces of the drawer part **32** may be defined by the separate plates **391**, **392** and **395**, which cover the components mounted on the drawer part **32**, and thus, the outer and inner appearances may be seen to be neat. The plates **391**, **392**, and **395** may be constituted by a plurality of plates and may be made of stainless steel to provide a more luxurious and clean appearance.

(92) As illustrated in the drawings, the front panel door part **31** and the drawer part **32** constituting the door **30** may be coupled to be separated from each other. Thus, assembling workability and serviceability may be improved through the separable structure of the front panel door part **31** and the drawer part **32**.

(93) A rear surface of the front panel door part **31** and a front surface of the drawer part **32** may be coupled to each other. When the front panel door part **31** and the drawer part **32** are coupled to each other, power for the elevation of the elevation device **80** may be provided. The driving device **40** for elevating the elevating device **80** may be disposed on the front panel door part **31**, and the front panel door part **31** and the drawer part **32** may be selectively connected to each other.

(94) Particularly, the driving part **40** provided in the front panel door part **31** may be configured to receive power from the power source and to transmit the power to the elevation part **80**. Thus, it is possible to remove the front panel door part **31** when the service of the driving part **40** is necessary and to take measures simply by replacing only the front panel door part **31**.

(95) The front panel door part **31** and the drawer part **32** may be coupled by a pair of door frames **316** provided on both sides. The door frame **316** includes a door coupling part **316a** extending upward and downward to be coupled to the front panel door part **31** and a drawer coupling part **316b** extending backward from a lower end of the door coupling part **316a**. The door coupling part **316a** may be coupled to the front panel door part **31** by a separate coupling member and may be coupled to one side of the front panel door part **31** by a simple coupling structure. Also, the drawer coupling part **316b** may be disposed on both sides of the drawer part **32** and adjacent to the draw-out rail **33**.

(96) The drawer coupling part **316b** may be inserted into the drawer part **32** to support the drawer part **32** in a state in which the door coupling part **316a** is coupled to the front panel door part **31**. Also, the drawer coupling part **316b** may be coupled to the drawer part **32** by a separate coupling member or may be coupled by a structure that mutually match the drawer coupling part **316b**.

(97) Also, a connection assembly **70** may be provided on the rear surface of the door **30** so that the driving part **40** and the elevation are **80** are connected to each other when the front panel door part **31** and the drawer part **32** are coupled. A drawer opening **35** through which a part of the elevation device **80** is exposed may be defined in a position corresponding to the connection assembly **70** on the front surface of the drawer part **32**.

(98) The front panel door part **31** may be configured to substantially open and close the storage space of the cabinet **10** and to define the front surface of the refrigerator **1**.

(99) The front panel door part **31** may have an outer appearance that is defined by an outer case **311** defining a front surface and a portion of a circumferential surface, a door liner **314** defining a rear surface, and an upper deco **312** and a lower deco **313** which respectively define top and bottom surfaces. Also, an insulating material **300** may be filled in the inside of the front panel door part **31** between an outer case **311** and a door liner **314**.

(100) Hereinafter, the front panel door part **31** and the driving assembly constituting the door **30** will be described in more detail with reference to the drawings.

(101) FIG. **5** is a rear view of the front panel door part. Also, FIG. **6** is a rear view illustrating a state in which a door cover of the front panel door part is removed. Also, FIG. **7** is an exploded perspective view of the front panel door part.

(102) A front surface of the front panel door part **31** may be defined by the outer plate **311**, and a

rear surface may be defined by the door liner **314**. Also, a driving device **40** for operating the elevation device **80** may be provided inside the front panel door part **31**. Although the driving device **40** may be disposed inside the front panel door part **31**, the driving device **40** but is not embedded in the insulating material **300** but is disposed inside the space defined by the door liner **314**. Then, the driving device **40** may be covered by the door cover **315** and thus may not be exposed to the outside.

(103) In detail, the insulating material **300** may be filled between the outer plate **311** and the door liner **314** to insulate the inside of the storage space **12**. Also, the door liner **314** may have a plurality of door recess parts that are recessed inward. The door recess parts may be defined in a shape corresponding to the shape of the elevation device **80** and may be recessed inside the door **30**.

(104) The door recess parts may include a motor recess part **314a**, a shaft recess part **314b**, a connector recess part **314c**, a lever recess part **314d**, and a screw recess part **314c**. Thus, the door recess parts may have shapes respectively corresponding to the constituents of the elevation device **80** so that the entire elevation device **80** is inserted into the inner space of the door **30**. Particularly, the lever recess part **314d** may include a rotation region of the lever **42** so that the lever **42** smoothly rotates during the operation of the driving device **40**.

(105) Also, the door recess part may include a light recess part **314f**. The light recess part **314f** may be recessed in an upper end of the rear surface of the door **30**. A door light **318** may be provided in the light recess part **314f**, and the inside of the door **30** may be illuminated by the door light **318**.

(106) In detail, the door light **318** may be defined to be long in the lateral direction from the left side to the right side of the rear surface of the door **30** and may be disposed at the uppermost position of the inner side regions of a gaskets **317** disposed along the rear surface of the door **30**.

(107) The door light **318** may include a plurality of LEDs **138c** and a light guide **318a** for guiding light emitted from the LEDs **138c** to the inside of the door **30**, i.e., the inside of the drawer part **32**.

(108) A plurality of the LEDs **138c** are disposed along the lower ends of the light guide **318a** and may be arranged to face the upper surface of the door **30** so that light is irradiated to the inner surfaces of the light guide **318a**.

(109) The light guide **318a** may have a shape corresponding to the light recess part **314f** and may have a curved surface. The light irradiated from the lower LEDs **138c** may be irradiated backward and downward to illuminate the inside of the drawer part **32**. The curved surface may be coated or surface-treated to reflect light and may be called a reflection surface.

(110) A light cover **318b** spaced apart from the front of the light guide **318a** may be disposed on the door light **318**. The light cover **318b** may also have a curved shape. Also, the light cover **318b** may be made of a transparent material capable of transmitting light. Thus, the light reflected from the light guide **318a** may be directed toward the inside of the drawer part **32**. Also, the light guide **318a** may guide an inflow of cool air to the inside of a space in which the driving device **40** is disposed to cool the driving device **40**.

(111) For this, the light cover **318b** may be exposed to the rear surface of the door **30**, and the lower surface of the curved surface of the light cover **318b** may be separated from the door cover **315** to define a space through which the cool air flows. Also, the air that cools the driving device **40** may be discharged through a door opening **315e** at a lower end of the door cover **315**. Thus, the cooling device may circulates the cool air circulating on the rear surface of the front panel door part **31** to cool the driving device **40** and assist the cooling circulation in the area around the drawer part **32**. Thus, the lower storage space **12** may be uniformly cooled.

(112) The door cover **315** may be configured to define an outer appearance of the rear surface of the front panel door part **31** and may be configured to cover the driving device **40** mounted on the front panel door part **31**. The door cover **315** may have a plate shape to cover the driving device **40** so that the door cover **315** is not exposed in the driving device **40** is mounted.

(113) The door cover **315** may have the cover recess part at a corresponding position to cover the driving device **40** from the rear side. The cover recess part may be recessed from the front surface

of the door cover **315**, i.e., the driving device **40**, and the rear surface of the door cover **315** may protrude toward the inside of the storage space. The cover recess part may include a motor recess part **315a**, a shaft recess part **315b**, and a lever recess part **315c**. Particularly, the lever recess part **315c** may include a rotation region of the lever **42** so that the lever **42** smoothly rotates during the operation of the driving device **40**.

(114) An upper end of the door cover **315** may be spaced apart from an upper end of the rear surface of the front panel door part **31**, and thus, the door light **318** may be exposed. Thus, a space for irradiating light to the inside of the drawer part **32** may be secured, and a space for supplying the cold air to the driving unit **40** may be provided.

(115) Also, a side cutout part **315d** may be defined in the left and right ends of the door cover **315**. The side cutout part **315d** may be a portion that exposes the supporter **319** to be coupled with the door frame **316** and may be defined inward in a shape corresponding to the supporter **319**.

(116) Also, a door opening **315e** may be defined in each of lower left and right sides of the door cover **315**. The door opening **315e** may be defined so that a portion of the connection assembly **70** passes through the door opening **315e** to protrude from the rear surface of the front panel door part **31**. Also, the door opening **315e** may have a corresponding shape at a position facing the drawer opening **35**. Thus, a portion of the connection assembly **70** exposed through the door opening **315e** when the front panel door part **31** and the drawer part **32** are coupled may be coupled to the elevation device **80** to transmit the power.

(117) The supporter **319** may be made of a metal material and fixedly mounted on the rear surface of the front panel door part **31**. Also, the supporter **319** may be exposed to both sides of the rear surface of the front panel door part **31** and may be firmly coupled to the door coupling part **316a** of the door frame **316** to maintain the state in which the front panel door part **31** is fixed and mounted on the drawer part **32**.

(118) The door opening **315e** may include a through-part **315g** and a guide part **315f**, and the through-part **315g** may be opened to allow the connection assembly **70** to be manipulated. The guide part **315f** may be opened along an operation path of the connection assembly **70** operating when the driving device **40** operates. Also, the door opening **315e** may be defined in a position facing the drawer opening **35** and may have the same shape as that of the drawer opening **35**.

(119) In detail, the through-part **315g** may be defined in a shape corresponding to the push part **741** of at least the connection assembly **70**. Thus, the user may manipulate the push part **741** exposed through the through-portion **315g** to selectively separate the connection assembly **70** and the elevation device **80** from each other.

(120) Also, the guide part **315f** may be opened to correspond to the rotation path of the connection member **73** rotating together with the rotation of the lever **42** rotated when the driving device **40** is driven. Thus, when the lever **42** and the connection member **73** rotate, the lever **42** and the connection member **73** may rotate without interfering with the door cover **315**.

(121) The door opening **315e** may pass through the rear portion of the front panel door part **31** to expose the connection assembly **70**. However, when the door opening **315e** is engaged with the drawer part **32**, the exposed portion may be covered.

(122) However, the door opening **315e** may be defined in a position farther forward than the cover recess part. Thus, when the push part **741** and the drawer part **32** are coupled to each other, the push part **741** and the front surface of the drawer part **32** may be slightly spaced apart from each other. Thus, the user may manipulate the push part **741** by inserting the hand into the space between the front panel door part **31** and the drawer part **32** in a state in which the front panel door part **31** and the drawer part **32** are coupled to each other.

(123) The door gasket **317** may be provided along the rear surface of the front panel door part **31**. When the door **30** is closed, the door gasket **317** may airtightly contact the front surface of the cabinet **10** in the state in which the door **30** is closed.

(124) The driving part **40** may be disposed inside the front panel door part **31** by being covered by

the door cover **315**. The driving device **40** may transmit the power to the elevation device **80** by the connection assembly **70** and also transmit the power to both sides of the elevation device **80** through the connection assemblies **70** disposed on both sides at the same time. Thus, the elevation device **80** may ascend and descend in the horizontal state at both left and right sides without being tilted or biased to one side under any situation.

(125) Hereinafter, the constituents of the driving device **40** will now be described in more detail with reference to the accompanying drawings.

(126) FIG. **8** is a perspective view of a door device according to a first embodiment. Also, FIG. **9** is an exploded perspective view of the driving device.

(127) As illustrated in the drawings, the driving device **40** may include a motor assembly **60**, a screw assembly **50** disposed on each of both sides of the motor assembly **60** and connected by a shaft **41**, a lever **42** connected to the screw assembly **50**, and the connection assembly **70**.

(128) In detail, the motor assembly **60** may be disposed at a center of both left and right sides of the front panel door part **31** and provide power for elevating the elevation device **80**. Also, the driving device **40** may allow both the screw assemblies **50** and the lever **42** to operate by the motor assembly including one driving motor **64**.

(129) Particularly, the motor assembly **60** may adjust magnitude of the decelerated and transmitted force through a combination of the plurality of gears. Also, a shaft **41** passing through the motor assembly **60** from the left to the right, i.e., in a horizontal direction may be disposed on an upper end of the motor assembly **60**, and the plurality of gears may be combined in the motor assembly **60** for rotation of the shaft **41**.

(130) Also, the motor assembly **60** may have a structure in which the driving motor **64** and the gears are arranged vertically to minimize a space recessed when the motor assembly **60** is mounted on the front panel door part **31**, in particular, a width in the left and right direction is widened, and a thickness in the front and rear direction is minimized. Also, the driving motor **64** constituting the motor assembly **60** may protrude toward the drawer part **32** to minimize a depth of the front panel door part **31** to secure insulation performance.

(131) The shaft **41** may pass through the motor assembly **60** in the transverse direction and be coupled to the screw assembly **50** disposed at both sides of the motor assembly **60** so that the power of the motor assembly **60** is simultaneously to the screw assembly (**50**). Thus, the shaft **41** may be called a power transmission member.

(132) For this, the shaft **41** may have a length such that both ends of the shaft **41** pass through the motor assembly **60** and are inserted into the screw assembly **50**. Also, a shaft driving gear **411** may be provided at a center of the shaft **41**. The shaft driving gear **411** may be coupled to the gears in the motor assembly **60** to rotate. Also, a shaft gear **412** may be disposed on each of both ends of the shaft **41**. The shaft gear **412** may have a structure that is coupled to the screw assembly **50**. The shaft gears **412** may have the same structure so that the same rotation force is applied to the shaft gears **412**. The screw assembly **50** may be transferred to the screw assembly **50** so that the screw assembly **50** operates simultaneously.

(133) The screw assemblies **50** may be disposed on both sides of the motor assembly **60**. The upper end of the screw assembly **50** may be connected to the shaft **41**. The shaft gear **412** is gear-coupled to transmit the power so that the screw **52** rotates, and a screw holder **56** moves along the screw **52**. Also, the lever **42** may be coupled to the screw holder **56** to allow the lever **42** to rotate according to the movement of the screw holder **56**.

(134) For this, the upper end of the screw assembly **50** may be oriented outward, and the lower end of the screw assembly **50** may be inclined inward. Here, the screw assemblies **50** on both sides may be symmetrical to each other with respect to the motor assembly **60**. Thus, the motor assembly **60** may be disposed between the screw assemblies **50** located on both sides of the screw assembly **50**. The screw assembly **50** disposed on both sides of the motor assembly **60** may be provided so that a distance between the screw assemblies **50** gradually increases from the upper end to the lower end.

(135) The screws **52** provided in the screw assembly **50** may be arranged in the same direction as the screw assembly **50**, and extension lines of the screws **52** on both the left and right sides may cross each other. Also, the screw holder **56** may move along the screw **52** according to the rotation of the screw **52**, and the lever **42** connected to the screw holder **56** may rotate along the connection assembly **70**. The screw assembly **50**, the lever **42**, and the connection assembly **70** may be symmetrical to each other so that the lever **42** simultaneously rotates at the same angle as the screw assembly **50** is driven.

(136) The lever **42** may connect the screw holder **56** to the connection assembly **70**. Thus, both ends of the lever **42** may be rotatably coupled to the screw holder **56** and the connection assembly **70**, respectively. Thus, when the screw holder **56** linearly moves, the lever **42** may be rotatable about the connection assembly **70**.

(137) The connection assemblies **70** disposed on both the left and right sides may be connected to each other by a connector bracket **43**, and the connection assembly **70** may be firmly supported on the front panel door part **31** to effectively transmit the rotation force to the elevation device **80**.

(138) Hereinafter, each constituent provided in the driving device **40** having the above-described structure will be described in more detail with reference to the drawings.

(139) FIG. **10** is a cross-sectional view of the screw assembly that is one component of the driving device. Also, FIG. **11** is an exploded perspective view of the screw assembly.

(140) The screw assembly **50** may be disposed on each of both left and right sides of the inside of the front panel door part **31**. Since the structure and the shape of the screw assembly **50** are different from each other only in the mounted position, only the screw assembly **50** will now be described.

(141) As illustrated in the drawings, the screw assembly **50** may include a housing **51**, a housing cover **55** for covering an opened upper surface of the housing **51**, a screw **52** provided inside the housing **51**, and a screw holder **56** which moves along the screw holder **56**.

(142) The housing **51** may define an outer appearance of the screw assembly **50** and provide a space in which a screw **52** and a screw holder **56** are accommodated. The opened upper surface of the housing cover **55** may be covered by the housing cover **55**.

(143) The housing **51** may be made of by bending a plate-shaped metal material, or may be made of a plastic material. The housing **51** may include the central portion **511** and the side part **512**. Also, a central portion **511** may be disposed at a position corresponding to the screw **52**, and at least a portion of the screw **52** may be accommodated in the central portion **511**. The central portion **51** may have a space in which the screw holder **56** coupled to the screw **52** moves vertically.

(144) The side part **512** may extend to be stepped at both sides of the central portion **511** and also extend from both side ends to both sides of the central portion **511** and be vertically bent to define both the side surfaces of the housing **51** and then be bent again inward from an end of both the side surfaces of the housing **51**.

(145) Thus, a space in which the screw **52** and the screw holder **56** are accommodated may be defined in the housing **51** by the side part **512**. Also, both side ends of the side part **512** may be bent outward, and a hole **512a** into which the coupling member is coupled may be defined in a state of being seated in the door recess part so that the housing **51** is fixed and mounted on the door liner **314**.

(146) The shape of the screw recess part **314e** disposed in the door liner **314** may have a stepped structure like the shape of the outer surface of the housing **51**. Thus, the door recess part and the outer surface of the housing **51** may be mutually coupled to each other so that the screw assembly **50** is firmly fixed without moving or being separated during the operation.

(147) A housing cutout part **513** may be disposed on the upper portion of the housing **51**. The housing cutout part **513** may be defined in a position corresponding to the position of the shaft gear **412** and the screw gear **53** disposed inside the housing **51**. The housing cutout part **513** may be

defined by cutting the shaft gear **412**. That is, the housing cutout part **513** may be cut so that the shaft gear **412** and the screw gear **53** do not interfere with each other when the shaft gear **412** and the screw gear **53** are coupled to each other.

(148) The screw **52** may be accommodated in the housing **51** and disposed at the central portion **511**. Also, the screw **52** may be disposed on an outer circumferential surface of the screw **52**. Thus, the screw holder **56** may move vertically along the screw **52** when the screw **52** rotates.

(149) A lower spacer **542** on which the screw **52** is rotatably supported may be disposed at a lower end of the screw **52**. A lower protrusion **523** protruding downward may be inserted into the screw **52**. The lower spacer may have the same structure as the bearing. Thus, the screw **52** may rotate in the state of being supported on the lower spacer **542**.

(150) The lower spacer **542** may be fixed and mounted on the lower cap **54**. The lower cap **54** may be mounted to cover the opened bottom surface of the housing **51** and define the bottom surface of the screw assembly **50**.

(151) The screw **52** may extend up to the upper end of the housing **51**, and the screw gear **53** and an upper spacer **541** may be mounted on the screw **52**.

(152) The screw gear **53** may be disposed on an upper end of the screw thread **521** and be integrally coupled to the screw **52** to rotate together with the screw **52**. Also, the screw gear **53** may be gear-coupled to the shaft **41** in the state of crossing the gear **412** mounted on the shaft **41**. Thus, the screw gear **53** and the shaft gear **412** may have the same shape as a bevel gear and provide a structure capable of transmitting the power in a crossing state.

(153) An upper protrusion **522** extending upward may be disposed on the upper end of the screw **52**. Also, the upper spacer **541** may be mounted to pass through the upper protrusion **522**. Also, the upper spacer **541** may be fixed to the inside of the housing cover **55** to rotatably support the upper end of the screw **52**.

(154) As described above, the upper and lower ends of the screw **52** may be rotatably supported by the upper spacer **541** and the lower spacer **542**. Also, the screw **52** may rotate by the power transmitted to the screw gear **53** by the shaft gear **412**, and the screw holder **56** may be elevated by the power.

(155) The screw holder **56** may include an elevation block **57**, a holder body **58**, and a holder cover **59**.

(156) The elevation block **57** may include a block body **571** having a block through-hole **571a** through which the screw **52** passes and a body coupling part **572** extending from the block body **571** in both lateral directions. The block body **571** may have a cylindrical shape, and the block through-hole **571a** may vertically pass through a center of the block body **571**. A screw corresponding to the screw thread **521** may be disposed on an inner circumferential surface of the block through-hole **571a**. Thus, when the screw **52** rotates, the screw **52** may move along the screw thread **521** to allow the elevation block **57** to vertically move.

(157) Also, a coupling hole **572a** may be defined in the body coupling part **572**. The coupling hole **572a** is defined in each of both sides of the block through-hole **571a**, and the screw may be coupled to allow the elevation block **57** to be coupled to the holder body **58** so that the elevation block **57** move together with the holder body **58**.

(158) The holder body **58** may be coupled to the elevation block **57** so as to be elevated together inside the housing **51**. The holder cover **59** may be coupled to one surface of the housing **51** exposed to the outside of the housing **51**.

(159) The holder body **58** may be hollow to provide a space. In particular, a block accommodation part **581** into the elevation block **57** is accommodated may be provided in a lower portion of the holder body **58**. The block accommodation part **581** may be opened backward and downward to communicate with the hollow. Thus, the elevation block **57** may be inserted and mounted from a lower side to an upper side of the holder body **58** and be disposed inside the block accommodation part **581**.

(160) A through-part **582** penetrated in the vertical direction may be defined above the block accommodation part **581**. The screw **52** may pass through the through-part **582** and may not contact a screw thread **521** of the screw **52**.

(161) A holder coupling part **583** to which the coupling member **572b** is coupled to couple the elevation block **57** to the holder body **58** may be disposed on each of both sides of an outer surface of the through-part **582**. The holder coupling part **583** may be disposed at a position corresponding to the body coupling part **572** and be integrally coupled to the coupling member **572** such as a screw at a position corresponding to each other when the elevation block **57** and the holder body **58** are coupled to each other.

(162) A side surface part of the holder body **58** may extend to both sides of the housing **51**. Also, a bearing unit **584** may be provided between each of both side surfaces of the holder body **58** and the inner surface of the housing **51**. The bearing unit **584** may include a bearing **584a**, which are vertically provided in plurality, and a retainer **584b** to which the bearing **584a** is rotatably mounted.

(163) Thus, the plurality of bearings **584a** may be mounted rotatably by the retainer **584b** and contact the side surface part of the holder body **58** and the inner surface of the housing **51** so as to be rolled. Of course, the bearing unit **584** is not limited to the above-described structure, and another structure capable of being rolled between the holder body **58** and the housing **51** may be also possible.

(164) The bearing unit **584** may be provided on each of both sides of the holder body **58**. Thus, the holder body **58** may smoothly ascend inside the housing **51**. Particularly, since the bearing units **584** on both sides are maintained in contact with the inner surface of the housing **51** during the elevation processes, the holder body **58** may be stably and smoothly elevated without moving. Also, the inner surface of the side part **512** contacting the bearing **584a** and both side surfaces of the holder body **58** may be recessed at positions corresponding to the bearing unit **584** to realize more stable rolling of the bearing **584a**.

(165) Although not shown, the screw assembly **50** may be provided with a shaft (not shown) for guiding the screws **52** on each of both sides of the screw **52** without the bearing unit **584**, and the screw holder **56** may move along the screw **52** without moving of the screw holder **56** through a structure in which the shaft extends from an upper end to a lower end of the housing **51** to pass through both sides of the screw holder **56**.

(166) The holder cover **59** may be coupled to one surface of the holder body **58**. The holder cover **59** may be coupled to the rear surface of the holder body **58** and be exposed to the outside through the opening of the housing **51**. Also, a holder protrusion **591** may protrude backward from the holder cover **59**. The holder protrusion **591** may pass through one end of the lever **42** and may have a circular cross-section so that the lever **42** rotate while passing through the lever **42**.

(167) Also, a protrusion restriction member **592** may be coupled to the end of the holder protrusion **591** passing through the lever. The protrusion restriction member **592** may be larger than the opening of the lever **42** through which the holder protrusion **591** passes. Also, the holder protrusion **591** may be inserted to pass through the lever **42**, and then, the protrusion restriction member **592** may be coupled to the end of the holder protrusion **591** by using a separate coupling member.

(168) In this embodiment, the screw holder **56** may have a structure in which the elevation block **57**, the holder body **58**, and the holder cover **59** are molded and coupled to each other. Here, the holder body **58** to which a load is directly applied may be made of a metal material. The elevation block **57** having a relatively complicated internal structure and the holder cover **59** to which the load is not applied may be formed by injection molding a plastic material.

(169) That is, since the elevation block **57** has to have a screw to move along the screw **52**, the structure may be complicated, and also, it may be difficult to mold the elevation block **567** having abrasion resistance and lubrication performance through engineering plastic injection molding. Also, the elevation block **57** to which a load is applied substantially when the elevation device **80** moves may have to have high strength. Thus, the holder body **58** may have a structure that is

capable of being molded by using a metal material through the extrusion. Also, the holder cover **59**, which is coupled to the lever **42** to generate friction during rotation of the lever **42**, may also be injection-molded using a plastic material. As described above, the elevation block **57**, the holder body **58**, and the holder cover **59**, which are made of different materials, may be coupled to each other and may be integrally elevated inside the housing **51** to match the respective structure and environment.

(170) Of course, the screw holder **56** may have a single structure, and a portion of the elevation block **57**, the holder body **58**, and the holder cover **59** may be integrally formed with each other.

(171) A housing cover **55** may be disposed on the upper end of the housing **51**. The housing cover **55** may have a structure that covers the top surface of the housing **51** and covers the screw gear **53** and the shaft gear **412** in the housing **51**.

(172) A bottom opening **551** through which the screw **52** passes may be defined in a bottom surface of the housing cover **55**. The upper portion of the screw **52** may be inserted into the housing cover **55** through the bottom opening **551**, and at least the screw gear **53** may be disposed inside the housing cover **55**. Also, the screw **52** may be fixed to the housing **51** and the housing cover **55** by the screw fixing member **531**, and the screw gear **53** may be fixed to always maintain the correct position.

(173) Also, an upper spacer mounting part **553** may be disposed on an inner top surface of the housing cover **55** to have a corresponding shape so that the upper spacer **541** is mounted. Thus, the housing cover **55** may be capable of rotatably supporting the upper end of the screw **52**.

(174) Also, a side opening **552** through which the shaft **41** is inserted may be defined in a side surface of the housing cover **55**. The side opening **552** may be opened so that the shaft gear **412** is disposed inside the housing cover **55**. Also, the shaft **41** may be fixed to the inside of the housing cover **55** by the shaft fixing member **612**, and the shaft gear **412** may be maintained in the state of being coupled to the screw gear **53** while being maintained at the correct position.

(175) Thus, the end of the shaft **41** and the screw **52** may be covered, and the shaft gear **412** and the shaft gear **412** may be covered when the housing cover **55** is mounted. Also, each of the shaft gear **412** and the screw gear **53** may be always disposed in a proper position to secure the power transmission through the shaft **41**.

(176) FIG. **12** is an exploded perspective view of the motor assembly that is one component of the driving part. Also, FIG. **13** is a view illustrating a coupling structure of the motor assembly and the driving shaft.

(177) As illustrated in the drawings, the motor assembly **60** may include a plurality of gears, a motor case **61**, and motor covers **62** and **63**.

(178) In detail, the driving motor **64** may provide power for elevating the elevation device **80** and may rotate forwardly and reversely. Thus, when an elevation signal of the elevation device **80** is inputted, the elevation device **80** may rotate forwardly and reversely to provide the power for elevating the elevating device **51**. Also, an input of a stop signal due to the load of the driving motor or the detection of the sensor may be stopped.

(179) The driving motor **64** may be fixed and mounted on a lower portion of the motor case **61**, and the rotation shaft of the driving motor **64** may pass through the motor case **61** to protrude to an opposite side. Also, the rotation shaft of the driving motor **64** may be provided with a first gear **651** to rotate when the driving motor **64** is driven.

(180) The driving motor **64** and the shaft **41** may be disposed on one side of the motor case **61**. A plurality of gears **651**, **652**, **653**, **654**, and **655** may be disposed on the opposite side of the motor case **61**. A second gear **652** engaged with the first gear **651** for transmitting and decelerating the power of the driving motor **64**, a third gear **653** engaged with the second gear **652**, a fourth gear engaged with third gear **653**, and a fifth gear **655** engaged with the fourth gear **654** may be disposed on one surface of the motor case **61**. Of course, the plurality of gears **651**, **652**, **653**, **654**, and **655** may be variously combined according to the reduction ratio and the magnitude of the

transmitted force. The plurality of gears **651**, **652**, **653**, **654**, and **655** may include at least a first gear coupled to the rotation shaft of the driving motor **64** and a fifth gear coupled to the shaft **41**.

(181) The fifth gear **655** may include a power transmission part **655a** and a power conversion part **655b**. The power transmission part **655a** is configured so as to be engaged with the fourth gear **654** in the form of a spur gear. Also, the power conversion part **655b** may be configured to be gear-coupled to the shaft driving gear **411** mounted on the shaft **41**.

(182) The shaft **41** may pass through the motor assembly **60** in a lateral direction and may extend in a direction perpendicular to the rotation axis of the driving motor **64** and the rotation axis of the fifth gear **655**. Also, the shaft driving gear **411** is disposed inside the motor assembly **60** and may be gear-coupled perpendicularly to the power conversion part **655b**. Thus, the power conversion part **655b** may have the same shape as a bevel gear so that power transmission to the shaft **41** is performed. The power conversion part **655b** and the shaft driving gear **411** may have other gear structures capable of transmitting the power.

(183) The motor case **61** may be provided with shaft fixing members **612** through which the shaft **41** passes. The shaft **41** passing through the shaft fixing member **612** may be provided with a shaft sleeve **414**. The shaft sleeve **414** may pass through the shaft fixing member **612**, and the shaft fixing member **612** may support the shaft sleeve **414**. Thus, the power conversion part **655b** and the shaft driving gear **411** may operate stably while being maintained in the engaged state therebetween without the movement of the shaft **41**.

(184) A rotation shaft **655c** may protrude from the rotation center of the fifth gear **655**. A plurality of guide protrusions **611** may protrude from the motor case **61** adjacent to the fifth gear **655**. Also, a gear restriction member **66** into which the rotation shaft **655c** and the guide protrusion **611** are inserted may be provided. The gear restriction member **66** may allow the fifth gear **655** to be maintained in the restricted state and include a rotation shaft hole **661** through which the rotation shaft **655c** passes. A guide hole **662** may be defined in a position corresponding to the guide protrusion **611**. Thus, in the state in which the gear restriction member **66** is mounted, separation and movement of the fifth gear **655** may be completely prevented to maintain the engaged and rotating state of the fifth gear **655** and the shaft gear **412**, thereby securing the power transmission to the shaft **41**.

(185) The motor covers **62** and **63** may include a front cover **62** and a rear cover **63** that respectively cover the front and rear surfaces of the motor case **61**. The front surface of the motor case **61** may face the door liner **314**, and the rear surface of the motor case **61** may face the door cover **315**.

(186) The front cover **62** may be coupled to the front surface of the motor case **61** and may cover the plurality of gears **651**, **652**, **653**, **654**, and **655** mounted on the front surface of the motor case **61**. The front cover **62** may be provided with a gear recess part **621**. Thus, the plurality of gears **651**, **652**, **653**, **654**, and **655** may be accommodated inside the gear recess part **621**, and the rotation axis may be fixed to realize the stable rotation. Also, the front cover **62** may further include a restriction member recess part **622** in which the gear restriction member **66** is accommodated.

(187) The rear cover **63** may be coupled to a rear surface of the motor case **61** and be configured to cover the driving motor **64** and a portion of the shaft **41** mounted on the rear surface of the motor case **61**.

(188) Thus, a shaft accommodation part **631** for covering the shaft driving gear **411** and the power conversion part **655b** of the fifth gear **655**, which are coupled to each other, may be provided in the rear surface of the motor case **61**. Also, a motor accommodation part **632** in which the driving motor **64** is accommodated may be provided. Also, the shaft fixing member **612** may be fixedly mounted on both left and right ends.

(189) A plurality of fixing parts **633** may protrude from both sides of the rear cover **63**. A plurality of vibration prevention members **67** may be press-fitted into the fixing part **633**, and the coupling member passing through the vibration prevention member **67** may be coupled to the door liner **314**.

to fix and mount the motor assembly **60**. The vibration prevention member **67** may be made of rubber or urethane to reduce vibration noise that is generated when the motor assembly **60** is driven.

(190) FIG. **14** is an exploded perspective illustrating a coupling structure of a connection assembly, which is one component of the driving device, and a lever.

(191) As illustrated in the drawing, the lever **42** may be configured to connect the screw assembly **50** to the connection assembly **70**.

(192) In details of the structure of the lever **42**, the lever **42** may be provided in a rod or bar shape having a predetermined width and may extend from the rotation axis of the connection assembly **70** to the holder protrusion **591** of the screw assembly **50**.

(193) In detail, the lever **42** may include a first extension part **421** connected to the connection assembly, a second extension part **423** connected to the screw holder **56**, and an intermediate portion **422** connecting the first extension part **421** to the second extension part **423**.

(194) The first extension part **421** and the second extension part **423** may be disposed parallel to each other, and the intermediate portion **422** may have an inclination. Also, the first extension part **421** may be further backward than the second extension part **423** by the inclination of the intermediate part **422**.

(195) The lever **42** may not be deformed or damaged even if a large amount of force is applied to the lever **42** due to the structure and shape of the bent lever **42**. Also, the lever **42** may be made of a metal material to realize the stable power transmission even when the elevation device **80** on which a heavy food is seated is elevated.

(196) Also, the inclination of the intermediate portion **422** may allow the lever **42** to be connected between the connection assembly **70** disposed relatively backward and the screw holder **56** disposed relatively forward.

(197) A first lever hole **424** may be defined in the first extension part **421** to be connected to the lever fixing member **75** of the connection assembly **70**. The first lever hole **424** may be formed in a polygonal shape corresponding to one side of the lever fixing member **75** and may be opened in a rectangular shape as illustrated in the drawing. The lever fixing member **75** may also rotate together when the lever **42** rotates.

(198) Also, the lever protrusion **425** may be disposed on the first extension part **421**. The lever protrusion **425** may be spaced apart from the first lever hole **424** and disposed toward the intermediate part **422**. The lever protrusion **425** may be configured to be coupled to the connection member **73** of the connection assembly **70**. That is, the rotation force of the lever **42** may be transmitted to the connection assembly **70** by the lever protrusion **425** together with the first lever hole **424**. Furthermore, the rotation force may be transmitted to the elevation device **80** to elevate the elevation device **80**.

(199) Also, a second lever hole **426** through which the holder protrusion **591** of the screw holder **56** is inserted may be defined in the second extension part **423**. The second lever hole **426** may have a size corresponding to the holder protrusion **591** and also may have a long hole shape in the extension direction of the second extension part **423** so that the holder protrusion **591** move as the screw holder **56** move vertically. Thus, the holder protrusion **591** may be disposed on the left end of the second lever hole **426** in a state in which the screw holder **56** is disposed at the lowest position, and as the screw holder **56** move upward, the protrusion **591** moves to the right side of the second lever hole **426** so that the lever **42** rotates.

(200) The connection assembly **70** may be provided at one end of the lever **42**, i.e., at a position corresponding to the first extension part **421**. A connection member **73** for connecting the lever **42** to the elevation device **80** may be rotatably mounted on the inside of the connection assembly **70**.

(201) The connection member **73** may be coupled to the lever fixing member **75** by the fixing shaft **77** and thus may rotate together with the rotation of the lever **42**. Also, the connection member **73** may be connected to the lever protrusion **425** and the scissors protrusion **841b** to transmit greater

force to the elevation device **80**, and thus, the elevation device **80** may be more effectively lifted. Thus, the elevation device **80** in the state in which the food is seated sufficiently while using only one of the driving motors **64** may be elevated, and a compact configuration may be realized.

(202) The connection assembly **70** may have an outer appearance defined by the connecting case **71** and the connecting cover **72**, and the lever fixing member **75** and the connection member **73** may be mounted on the connecting case **71**.

(203) Hereinafter a structure of the connection assembly **70** will be described in more detail.

(204) FIG. **15** is an exploded perspective view of the connection assembly when viewed in one direction. Also, FIG. **16** is an exploded perspective view of the connection assembly when viewed in the other direction. Also, FIGS. **17** and **18** are views illustrating an operation state of the connection assembly.

(205) Referring to the drawings, the connection assembly **70** may include the connecting case **71**, the connecting cover **72**, and the connection member **73**, the push member **74**, the lever fixing member **75**, and the clastic member **76**.

(206) In detail, the connecting case **71** may be opened on one side and includes a space **711** for accommodating the lever fixing member **75**, the connection member **73**, the push member **74**, and a portion of the lever **42**. Also, a through-hole **712** may be defined in the space **711**. An external fixing member **78** may be provided on the outer surface of the connecting case **71** corresponding to the through-hole **712**.

(207) The lever fixing member **75** may include an elastic support part **751** and a through-protrusion **752**. The elastic support part **751** may be accommodated in the space inside the connecting case **71** and define a surface capable of supporting one end of the elastic member **76**. Also, the through-protrusion **752** may be disposed on a center of the clastic support part **751** and extend to sequentially pass through the first lever hole **424** and the through-hole **712** of the lever. The through-protrusion **752** may have a rectangular cross-sectional shape. The through-protrusion **752** may be inserted into a fixing groove defined in the external fixing member **78**.

(208) A shaft insertion part **752a** into which the fixing shaft **77** is inserted may be provided inside the through-protrusion **752**. The fixing shaft **77** and the shaft insertion part **752a** may have corresponding shapes and may have a rectangular cross-section like the through-protrusions **752**. Thus, slippage may not occur during the rotation of the lever **42**, stable rotation force may be transmitted to the lever fixing member **75**.

(209) The fixing shaft **77** may be inserted into the shaft insertion part **752a** of the through-protrusion **752** after passing through the first connection part **731** of the connection member **73**. Also, the fixing shaft **77** may be inserted into the shaft insertion part **752a**. Also, the coupling member **771** and **772** may be coupled to both ends of the fixing shaft **77**. The lever fixing member **75**, the external fixing member **78**, and the connection member **73** may be coupled to the fixing shaft **77** through the coupling of the coupling members **771** and **772**. Thus, when the lever fixing member **75** rotates by the rotation of the lever **42**, the connection member **73** connected by the fixing shaft **77** may also rotate together.

(210) The elastic member **76** may be provided between the connection member **73** and the lever fixing member **75**. The elastic member **76** may be compressed when the connection member **73** moves. In detail, the elastic member **76** may have a coil spring structure and have one end supported by the elastic support part **751** and the other end supported by the connection support part **734** of the connection member **73**.

(211) The connection member **73** may move in the front-rear direction within the space of the connecting case **71**. Here, the connection member **73** may have a structure that is inserted into or protrudes to the space by the guide of the fixing shaft **77**.

(212) In details of the structure of the connection member **73**, the connection member **73** may include a first connection part **731** which passes through the fixing shaft **77** and is concentric with the rotation axis of the lever **42**, a second connection part **732** which is spaced from the first

connection part **731** and into which the lever protrusion **425** is inserted, and a connection member **73** connecting the first connection part **731** to the second connection part **732**.

(213) The first connection part **731** may have a hollow cylindrical shape. The first connection part **731** may have a first hollow part **731a** into which the fixing shaft **77** is inserted, a second hollow part **71b** which has a diameter greater than that of the first hollow part **731a** and to which the coupling member **771** coupled to the fixing shaft **77** is disposed, and a third hollow part **731c** which has a diameter greater than that of the second hollow part **731b** and into which the rod protrusion **841a** of the elevation device **80** is inserted.

(214) The first hollow part **731a** may have a rectangular cross-section like the fixing shaft **77**, and the second hollow part **731b** may have a circular cross-section. Also, at least a portion of the third hollow part **731c** may have a groove shape corresponding to a rotation trajectory of an end of the rod protrusion **841a** so that the rod protrusion **841a** of the elevation device **80** is inserted, and when the elevation device **80** rotates, the rod protrusion **841a** is hooked after rotating at a predetermined angle. As illustrated in FIG. **21**, the rod protrusion **841a** may have a planar shape on both sides thereof and be hooked with an stepped inner portion of the third hollow part **731c** so that the rod protrusion **841a** is hooked inside the third hollow part **731c**.

(215) Also, a connection support part **734** protruding outward by a predetermined width may be disposed on one side of the first connection part **731**. The end of the elastic member **76** may contact the connection support part **734**, and the end of the first connection part **731** may contact the connection support part **734**. The connection support part **734** may protrude outward to support one end of the elastic member **76**, and one end of the first connection part **731** may be inserted into the elastic member **76** to prevent the elastic member **76** from being separated.

(216) The connection support part **734** may be larger than the size of the through-hole **742** defined in the push member **74** to maintain the state in which the connection support part **734** is in close contact with the rear surface of the push member **74**. Thus, the connection support part **734** and the push member **74** may move together when the push member **74** is pressed or when the elastic member **76** returns to the initial position.

(217) The second connection part **732** may be disposed at a position spaced apart from the first connection part **731** by the connection member **73**. The second connection part **732** may have a cylindrical shape having a hollow **732a** penetrated in the front and rear direction. The lever protrusion **425** may be inserted into one side of the second connection part **732**, and the scissors protrusion **841b** may be inserted into the other side of the second connection part **732**. Here, the lever protrusion **425** and the scissors protrusion **841b** may have the same outer diameter and correspond to the inner diameter of the second connection part **732**.

(218) The connection member **73** may be disposed so that the rod protrusion **841a** and the scissors protrusion **841b** of the elevation device **80** are respectively inserted into the first connection part **731** and the second connection part **732**. As the second connection part **732** move farther away from the first connection part **731**, the elevation device **80** may be easily elevated. However, when the first connection part **731** and the second connection part **732** are spaced a set distance or more from each other, the moving trajectory of the lever protrusion **425** and the scissors protrusion **841b**, which are inserted into the second connection part **732**, may extend up to a high height on the rear surface of the front panel door part **31** and the front surface of the drawer part. Thus, the opened trajectory may be exposed to deteriorate the outer appearance. Thus, the position of the second connection part **732** may be determined by the length of the connection member **73**. Also, the second connection part **732** may be disposed at a height at which the rotation trajectory is not exposed, i.e., a position higher than the upper end of the elevation device **80**.

(219) The push member **74** may be provided inside the connecting case **71** and may be exposed through the opening **721** of the connecting cover **72** so that the push member **68** is pressed by the user. The push member **74** may include a push part **741** exposed through an opening **721** of the connecting cover **72** and a push guide part **744** extending along a portion of the circumference of

the push part **741**.

(220) A through-hole **742** through which the first connection part **731** passes may be defined in the push part **741**. The through-hole **742** may be larger than the outer diameter of the first connection part **731** and slightly smaller than the outer diameter of the connection support part **734**. Thus, when the push part **741** may be pushed to move the push member **74**, the first connection member **73** contacting the push member **74** may also move together to selectively connect the connection member **73** to the elevation device **80**.

(221) Also, the circumference of the push part **741** may extend toward the connecting case **71** and then be bent outward to provide a push flange **743**. Thus, the push flange **743** may interfere with the opening **721** of the connecting cover **72** so that the push member **74** is restricted by the connecting cover **72** without being separated. For this, the opening of the connecting cover **72** may have a stepped part **722**, and the push flange **743** may be accommodated into the rear surface of the stepped part **722**.

(222) The push guide part **744** may be disposed on one side of a circumference of the push part **741**. The push guide part **744** includes a guide surface **744a** extending along the circumference of the push part **741** and contacting an inner surface of the connecting case **71** and a guide boss disposed on each of both sides of a guide surface **744a**. Also, the guide boss **744b** may be penetrated by a guide post **713** extending from the recessed bottom surface of the connecting case **71**.

(223) Thus, when the push member **74** move forward and backward, the guide surface **744a** may maintains the contact with the inner surface of the connecting case **71**, and the guide boss **744b** may move along the guide post **713** on each of both sides. Thus, the push member **74** may move forward and backward in the stable state without moving.

(224) The connecting cover **72** may be mounted on the opened front side of the connecting case **71**, and an opening **721** may be defined to expose the push part **741**. The connecting cover **72** may be firmly fixed to the connecting case **71** by the coupling member. Thus, the configuration of the connecting case **71** may be maintained in the mounted state.

(225) The connecting case **71**, the push member **74**, and a portion of the connecting cover **72** may be opened by cutting the connection member **73** by a rotational trajectory. Thus, the connection member **73** may be prevented from interfering with the connecting case **71**, the push member **74**, and the connecting cover **72** when the connection member **73** rotates.

(226) In this structure, the user may manipulate the push member **74** of the connection assembly **70** to selectively couple and separate the connection assembly **70** to and from the elevation device **80**.

(227) Hereinafter, a structure of the drawer part **32** coupled to the front panel door part **31** will now be described in more detail with reference to the accompanying drawings.

(228) FIG. **19** is an exploded perspective view of the drawer part.

(229) As illustrated in the drawings, the drawer part **32** may include a drawer body **38** defining an entire shape of the drawer part **32**, an elevation device **80** provided in the drawer body **38** to elevate the container and food, and a plurality of plates **391**, **392**, and **393** defining an outer appearance of the drawer part **32**.

(230) In more detail, the drawer body **38** may be injection-molded by using a plastic material and define an entire shape of the drawer part **32**. The drawer body **38** may have a basket shape having an opened top surface to define a food storage space therein. An inclined surface **321** may be disposed on a rear surface of the drawer body **38**. Thus, an interference with the machine room **3** may not occur.

(231) The door frames **316** may be mounted on both sides of the drawer part **32**. The door frame **316** may be coupled to the lower frame of each of both sides of the bottom surface or both left and right surfaces of the drawer part **32**. In the state in which the door frame **316** and the drawer part **32** are coupled to each other, the drawer part **32** and the front panel door part **31** may be integrally coupled to be inserted and withdrawn.

(232) The door frame **316** may be separated from the drawer part **32**, and then the connection assembly **70** may operate to separate the front panel door part **31** from the drawer part **32** in order to separate the front panel door part **31** from the drawer part **32**. The door frame **316** and the drawer part **32** may be coupled to each other by a separate coupling member or a coupling structure between the door frame **316** and the drawer part **32**.

(233) The draw-out rack **34** may be disposed on each of both the sides of the bottom surface of the drawer part **32**. The drawer part **32** may be inserted and withdrawn forward and backward by the draw-out rack **34**. In detail, in the state in which the drawer part **32** is mounted on the cabinet **10**, at least a portion is disposed in the storage space. Also, the draw-out rack **34** may be coupled to a pinion gear **141** disposed on the bottom surface of the storage space. Thus, when the draw-out motor **14** is driven, the pinion gear **141** may rotate to allow the draw-out rack **34** to move, and the door **30** may be inserted and withdrawn.

(234) The door **30** may not be automatically inserted and withdrawn. That is, the user may push or pull the door **30** to be inserted and withdrawn. Here, the draw-out rack **34** may be omitted, and thus, the insertion and withdrawal may be performed through only the draw-out rail **33**.

(235) A rail mounting part **382** on which the draw-out rail **33** for guiding the insertion and withdrawal of the drawer body **38** is mounted may be disposed on a lower portion of each of both the side surfaces of the drawer body **38**. The rail mounting part **382** may extend from a front end to a rear end and provide a space in which the draw-out rail **33** is accommodated. The draw-out rail **33** may be a rail that extends in multistage. The draw-out rail **33** may have one end fixed to the storage space inside the cabinet **10** and the other end fixed to the rail mounting part **382** to more stably realize insertion and the withdrawal of the door **30**.

(236) Also, the plurality of plates **391**, **392**, and **393** made of a plate-shaped metal material such as stainless steel to define at least portions of the inside and outside of the drawer body **38** may be provided on the drawer body **38**.

(237) In detail, the outer side plate **391** may be disposed on each of both left and right surfaces of the outside of the drawer body **38**. The outer side plate **391** may be mounted on each of both the left and right surfaces of the drawer body **38** to define an outer appearance of each of both the side surfaces. Particularly, the constituents such as the door frame **316** and the draw-out rail **33**, which are mounted on both the sides of the drawer body **38** may not be exposed to the outside.

(238) A plurality of reinforcement ribs **384** may cross each other in vertical and horizontal directions on both outer surfaces of the drawer body **38**. The reinforcement ribs **384** may reinforce the strength of the drawer body **38** itself so that the drawer body **38** is more rigidly shaped relative to the weight of the door, which increases by providing the driving device and the elevation. Also, the reinforcement ribs **384** may support the outer side plates **391** mounted on both side surfaces, and thus the outer appearance of the drawer part **32** may be firmly maintained.

(239) An inner side plate **392** may be disposed on each of both left and right surfaces of the inside of the drawer body **38**. The inner side plate **392** may be mounted on each of both the side surfaces of the drawer body **38** to define both the left and right surfaces of the inside thereof.

(240) The inner plate **395** may be constituted by a front part **395a**, a bottom surface part **395b**, and a rear surface part **395c**, which have sizes correspond to the front surface, the bottom surface, and the rear surface of the inside of the drawer body **38**. The inner plate **395** may be provided by bending the plate-shaped stainless material so that the inner plate **395** defines the inner surface of the remaining portion except for both the left and right surfaces of the drawer body **38**. Also, both left and right ends of the inner plate **395** may contact the inner side plate **392**. The front part **395a**, the bottom surface part **395b**, and the rear surface part **395c** constituting the inner plate **395** may be separately provided and then coupled to or contact each other.

(241) The entire inner surfaces of the drawer body **38** may be defined by the inner side plate **392** and the inner plate **395**, and the inner surface of the drawer body **38** may provide texture of the metal. Thus, the storage space within the drawer part **32** may have a metal texture on the whole,

and the foods accommodated in the drawer part **32** may be more uniformly cooled and thus stored at a low temperature in the more uniform region. In addition, visually excellent cooling performance and storage performance may be provided to the user.

(242) The drawer cover **37** may include a cover front part **371** that partitions the inside of the drawer body **38** into a front space **S1** and a rear space **S2** and a cover top surface part **372** bent from an upper end of the cover front part **371** to cover a top surface of the rear space **S2**.

(243) That is, when the drawer cover **37** is mounted, only the front space **S1**, in which the elevation device **80** is disposed, may be exposed in the drawer body **39**, and the rear space **S2** may be covered by drawer cover **37**.

(244) The elevation **80** may be disposed in the drawer body **38**. The elevation device **80** may be connected to the connection assembly **70** and may be vertically movable. The left and right sides of the elevation device **80** may be elevated uniformly.

(245) A drawer opening **35** may be defined in the lower part of the front surface of the drawer part **32** for coupling the elevation device **80** to the connection assembly **70**. The drawer opening **35** may provide a passage through which the connection member **73** is inserted to be coupled to the elevation device. Also, the drawer opening **35** may have an opening shape along the rotation path of the connection member **73** when the connection member **73** rotates to allow the connection member **73** to rotate, and thus, the stable rotation may be achieved without the interference.

(246) The elevation device **80** may be provided as a scissors type so that the elevation device is folded in a descending state and unfolded in an ascending state. Thus, the container or food seated on the upper surface may be elevated.

(247) The elevation device **80** may be provided with a support plate **81**, and the support plate **81** may provide a seating surface on which the container **36** or food is seated.

(248) Hereinafter, a connection structure between the connection assembly **70** and the elevation device will be described.

(249) FIG. **20** is an exploded perspective view illustrating a coupling relationship between the drawer part and the connection assembly. Also, FIG. **21** is an enlarged view illustrating a portion A of FIG. **20**.

(250) As illustrated in the drawings, the drawer opening **35** may be defined in the right and left sides of the lower front of the drawer part **32**. The shape of the drawer opening **35** on each of both sides of the right and left sides may be symmetrical to each other, and the rod protrusion **841a** of the elevation device **80** and the scissors protrusion **841b** may be exposed through the drawer opening **35**. That is, the drawer opening **35** may be opened at a position corresponding to the rod protrusion **841a** of the elevation device **80** and the scissors protrusion **841b**.

(251) The drawer opening **35** may include a central portion **351** and a trajectory portion **352**. The central portion **351** may be disposed at a position corresponding to the rod protrusion **841a** of the elevation device **80** and may have a size such that the first connection part **731** of the connection member **73** is inserted. Also, the trajectory portion **352** may be connected to the central portion **351** and may be opened in a shape corresponding to the trajectory in which the second connection part **732** of the connection member **73** move to rotate. Thus, the rod protrusion **841a** of the elevation device **80** may rotate on the central portion **351** while the scissors protrusion **841b** of the elevation device **80** rotates along the trajectory portion **352**. That is, the scissors protrusion **841b** and the second connection part **732** may be disposed inside the central portion **351** and the trajectory portion **352** when the elevation device **80** moves vertically.

(252) The height of the drawer opening **35** may be lower than the upper end of the elevation device **80**, i.e., the upper surface of the support plate **81**. Thus, the drawer opening **35** may be prevented from being seen from the inside of the drawer part **32** in any state in the state in which the elevation device **80** is mounted.

(253) The rod protrusion **841a** and the scissors protrusion **841b** of the elevation device **80** may be exposed through the drawer opening **35** while the elevation device **80** is mounted inside the drawer

part **32**. Also, in the state in which the sub door **30** is coupled, the connection member **73** of the connection assembly **70** may be inserted through the inside of the drawer opening **35** so as to be coupled to the rod protrusion **841a** of the elevation device **80** and the scissors protrusion **841b**. (254) The connection assembly **70** may be provided on each of both right and left sides of the drawer part **32** and may have a shape symmetrical to each other. The selective separation of the elevation device **80** and the connection assembly **70** may be enabled through the manipulation of the push member **74**.

(255) The circumference of the support plate **81** may protrude upward so that the container **36** or food is stably mounted. Also, the circumference of the support plate **81** may extend downward. Thus, the remaining constituents of the elevation device **80** may be accommodated below the support plate **81**, and the covered and clean outer appearance may be realized by the circumference of the support plate **81**.

(256) In addition, the support plate **81** may have a size and a shape corresponding to the front space to prevent foreign matters from being introduced into the elevation device **80** provided below the front space **S1**, and also, to fundamentally prevent safety accidents from occurring by blocking the access to the elevation device **80**.

(257) Hereinafter, constituents of the elevation device **80** will be described in more detail.

(258) FIG. **22** is a front view of an elevation device according to the first embodiment. Also, FIG. **23** is an exploded perspective view illustrating a state in which a support plate is separated from the elevation device.

(259) As illustrated in the drawings, when the elevation device is viewed from an upper side, the elevation device may have a rectangular shape having a size corresponding to that of a front space of the drawer part **32** and have a predetermined thickness.

(260) The elevation device **80** may be mounted on the bottom surface of the inner side of the drawer part **32** and may be detachably installed on the inside of the drawer part **32**. That is, the coupling between the elevation device **80** and the connection assembly **70** may be released in the state in which the elevation device **80** is mounted on the drawer part **32**. Thus, in the state in which the driving device **40** is mounted on the front panel door part **31**, only the elevation device may be separated from the inside of the drawer part **32**.

(261) The elevation device **80** may include an upper frame **82** and a lower frame **83** as a whole and a lifting assembly **84** disposed between the upper frame **82** and the lower frame **83**.

(262) An outer appearance of the elevation device **80** may be defined by the upper frame **82** and the lower frame **83**. Each of the upper frame **82** and the lower frame **83** may have a rectangular frame shape and also have a predetermined width and a predetermined thickness.

(263) A lower end of the upper frame **82** and an upper end of the lower frame **83** may have shapes corresponding to each other. In a state in which the upper frame **82** is disposed at the lowermost position, the lower end of the upper frame **82** and the upper end of the lower frame **83** may be coupled or contact each other.

(264) Also, the lifting assembly **84** may be disposed between the upper frame **82** and the lower frame **83**. The lifting assembly **84** may be connected to the connection assembly **70**. The lifting assembly **84** may be folded or unfolded according to the rotation of the lever **42** to allow the upper frame **82** to be elevated.

(265) When the upper frame **82** is disposed at the lowermost position, the lifting assembly **84** may be accommodated between the upper frame **82** and the lower frame **83**. When the support plate **81** is mounted, the upper frame **82** and the lower frame **83** as well as the lifting assembly **84** may not be exposed to the outside.

(266) Here, the scissors protrusion **841b** and the rod protrusion **841a**, which transmit the rotation force, of the lifting assembly **84** may protrude forward through the upper frame **82** and the lower frame **83**. When the elevation device **80** is mounted, the scissors protrusion **841b** and the rod protrusion **841a** may be coupled to the connection assembly **70**.

(267) The support plate **81** may be one component of the elevation device **80** and may be seated on the upper frame **82** to define the top surface of the elevation device **80**. Also, the container **36** or food may be seated on the support plate **81**. Thus, the upper frame **82** and the support plate **81** may be elevated together with the elevation device **80** to elevate the container **36** or food.

(268) In details of the support plate **81**, the support plate **81** may include a support surface **811** having a plate shape and an edge **812** disposed around the support surface **811**.

(269) The support surface **811** may have disposed at a position lower than that of the edge **812**. The support surface **811** may have a size and shape corresponding to those of the bottom surface of the container **36**. Thus, when the container **36** is seated on the support plate **81**, the container **36** may be inserted into a space that is defined by the edge **812**, and the bottom surface of the container **36** may be supported by the support surface **811**. Thus, while the elevation device **80** is elevated, the container **36** may be maintained in a stably accommodated state.

(270) The edge **812** may have a predetermined width, and an outer surface of the edge **812** may extend downward. The edge **812** may extend to contact a circumferential surface of the upper frame **82**. Thus, in the state in which the support plate **81** is mounted, the circumferential surface of the upper frame **82** may be covered by the edge **812**. Also, when viewed from an upper side, the upper frame **82** made of a metal material may not be exposed to the outside. That is, only the support plate **81** made of a plastic material may be exposed to the outside through the front space.

(271) A handle part **813** may be disposed on an edge **812**, which defines both left and right side surfaces of the support plate **81**, of the edge **812**. A stepped part **814** that is stepped inward may be disposed on each of both sides of the edge **812** on which the handle part **813** is disposed, and the handle part **813** may be recessed at a center of the stepped part **814**. Thus, the user may lift the support plate **81** by putting his hand into the stepped part **814** and the handle part **813** in the state in which the elevation device **80** is accommodated in the drawer part **32** and also may detach the support plate **81** from the support plate **82**.

(272) A cutoff part **815** may be defined in an edge portion of the support plate **81** so that the support plate **81** is easily mounted, and a portion of the edge of the upper frame **82** may be exposed. Also, a groove **816** recessed to prevent the rod protrusion **841a** from interfering may be further defined in the front surface of the support plate **81**.

(273) Hereinafter, structures of the upper frame, the lower frame **83**, and the lifting assembly **84**, which constitute the elevation device **80**, will be described in more detail with reference to the accompanying drawings.

(274) FIG. **24** is a perspective view of the elevation device. Also, FIG. **25** is an exploded perspective view of the elevation device in a state in which the lifting assembly that is one component of the elevation device is unfolded. Also, FIG. **26** is an exploded perspective view of the elevation device in a state in which the lifting assembly is folded. Also, FIG. **27** is a perspective view of the upper frame that is one component of the elevation device.

(275) As illustrated in the drawings, the upper frame **82** may have a square frame shape corresponding to the size of the inner front space **S1** of the drawer part **32** and may be configured to mount the support plate **81** on the top surface thereof.

(276) The upper frame **82** may substantially support the food or container **36** together with the support plate **81** so as to be elevated together. Also, the upper frame **82** may generally defines a frame part **821** which defines a circumferential shape of the upper frame **82** and a partition part **822** for partitioning the space inside the frame part **821** into left and right sides.

(277) Since the frame part **821** and the partition part **822** define an outer frame and support the support plate **81**, high strength may be required, and thus, the frame part **821** and the partition part **822** may be made of a metal and may have shape in which both ends are bent to increase the strength and prevent deformation.

(278) In detail, each of the frame part **821** and the partition part **822** may be formed of a plate-shaped metal material, and both ends of each of the frame part **821** and the partition part **822** may

be bent to provide a bent portion. Then, grooves, holes, or cut portions that are necessary for a connection relationship with other constituents may be formed to be coupled to each other.

Particularly, the frame part **821** may be configured so that the constituents defining both left and right surfaces of the upper frame **82** are formed and then coupled to each other. Also, the partition part **822** may be coupled in a state in which the frame part **821** is assembled. Spaces **823** and **824** may be defined in both sides of the upper frame **82** with respect to the partition part **822**.

(279) Edges **821a**, **821b**, **831a**, and **831b** may be provided on all of the upper frame **82** and the lower frame **83**. In the state in which the upper frame **82** is disposed at the lowermost position, lower ends of the edges **821a** and **821b** of the upper frame **82** and the edges **831a** and **831b** of the lower frame **83** may contact each other to define spaces. Also, the lifting assembly **84** may be accommodated in a space between the upper frame **82** and the lower frame **83**.

(280) Also, a slide guide **825** may be disposed on each of both sides of the inner side of the upper frame **82** to accommodate the end of the lifting assembly **84** and guide the movement of the lifting assembly **84**. The slide guides **825** may be disposed on both sides of the partition part **822**. Also, the scissors assemblies **84** having the same structure may be disposed on both sides with respect to the partition part **822**.

(281) The slide guide **825** may be separately molded by using a plastic material having excellent abrasion resistance and lubrication performance and mounted on the upper frame **82**. Also, a long hole **825a** through which the sliding shaft **842** of the lifting assembly passes may be defined in the slide guide **825**, and the sliding shaft **842** may move along the slide guide **825**. Also, a sliding surface **825b** having a predetermined width may be further disposed along the circumference of the long hole **825a**, and the sliding shaft **842** may be more stably supported by the sliding surface **825b** so that the lifting assembly **84** is more smoothly folded or unfolded.

(282) The frame part **821** may include vertically curved edges **821a** and **821b** along the circumference thereof. The edges **821a** and **821b** may be disposed on the inner side and the outer side of the frame part **821**, respectively. Also, the slide guide **825** may be disposed on the edge **821b** inside the frame part **821**.

(283) Also, an edge groove **821d** may be defined in the outer edge **821a** of the frame part **821**. The edge groove **821d** may be a groove in which the scissors protrusion **841b** of the elevation device **80** is accommodated in the state in which the elevation device **80** completely descends and be defined in a position corresponding to the scissors protrusion **841b** at the end of the edge **821a**. When the upper frame **82** completely descends to contact the lower frame **83**, the upper frame **82** may contact the edge groove and **821d** defined in the lower frame **83** to provide a complete hole shape so that the scissors protrusion **841b** passes therethrough.

(284) The edge groove **821c** may be defined in a position corresponding to the rod protrusion **841a** according to the size or position of the rod protrusion **841a**. The edge grooves **821c** and **821d** and the rod protrusion **841a** and the scissors protrusion **841b** may be disposed adjacent to the left and right ends of the elevation device **80** and may be exposed through the drawer opening **35**.

(285) Also, a coupling groove **821f** and a coupling end **821e** may be provided on both ends of the upper frame **82**, which correspond to both surfaces of the drawer part **32**. The coupling groove **821f** and the coupling end **821e** may be provided on the extending end of the edge **821a**. Also, the coupling end **821e** may be provided to protrude downward, and the coupling groove **821f** may be defined to be recessed upward. The coupling groove **821f** and the coupling end **821e** may be disposed on both the upper frame **82** and the lower frame **83**. When the upper frame **82** completely move downward, the upper frame **82** may be coupled to the coupling end **821e** and the coupling groove **821f**.

(286) Also, a scissors fixing member **826** may be provided at each of both ends of the inner space of the frame part **821**. The scissors fixing member **826** may fix the rotation shaft **847** of the lifting assembly **84**, and a pair of scissors fixing members **826** may be provided at both ends.

(287) The scissors fixing member **826** may be provided inside the accommodation space **830** of the

frame part **821** corresponding to the coupling end **821e**. That is, the coupling end **821e** may have a size corresponding to one surface of the scissors fixing member **826** so that the scissors fixing member **826** is fixed and mounted. Thus, the coupling end **821e** may provide a surface that is coupled to the scissors fixing member **826** by a separate coupling member such as a screw.

(288) The scissors fixing member **826** may also be made of an engineering plastic material having abrasion resistance due to continuous friction with the rotation shaft **847**. Also, the scissors fixing member **826** may have a through-hole **843b** through which the rotation shaft **847** passes.

(289) A plurality of scissors fixing members **826** may be provided on both ends of the frame part **821** so that both ends of the rotation shaft **847** are rotatably fixed. Thus, the rotation shaft **847** may be stably fixed to allow the lifting assembly **84** to be smoothly folded and unfolded.

(290) The lower frame **83** may have the same structure as that of the upper frame **85** but only in the direction. The lower frame **83** may include a frame part **831** and a partition part **832** to define spaces **833** and **834** in which the scissors assemblies **84** are respectively installed.

(291) Also, the slide guide **835** may be provided on the inner edge **831b** of the lower frame **83**. The slide guide **835** may have the same structure and shape as the slide guide **825** of the upper frame **82**. Also, a second sliding shaft **846** of the lifting assembly **84** may pass through the slide guide **835** of the lower frame **83** to move along the slide guide **835**.

(292) Also, a first frame groove **831c** and a second frame groove **831d** may be defined in the outer edge **821a** of the lower frame **83**. The first frame groove **831c** and the second frame groove **831d** may be defined in positions corresponding to the frame groove **821d** defined in the upper frame **82**. In the state in which the upper frame **82** and the lower frame **83** contact each other, the scissors protrusion **841b** and the rod protrusion **841a** may pass through the upper frame **82** and the lower frame **83** to protrude outward.

(293) The coupling end **831e** and the coupling groove **831f**, which match the end of the upper frame **82**, may be provided on both side surfaces of the frame part **831** of the lower frame **83**. The coupling end **831e** and the coupling groove **831f** of the lower frame **83** disposed on the lower frame **83** may be disposed at positions facing the coupling groove **821f** the coupling end **821e** and have shape corresponding to those of the coupling groove **821f** the coupling end **821e** so that the coupling end **831e** and the coupling groove **831f** are respectively coupled to the coupling groove **821f** the coupling end **821e**. That is, the coupling end **821e** of the upper frame **82** and the coupling groove **831f** of the lower frame **83** may be disposed at positions facing each other, and the coupling groove **821f** of the upper frame **82** and the coupling end **831e** of the lower frame **83** may be disposed at positions facing each other. Thus, when the upper frame **82** and the lower frame **83** are coupled to each other, a lower end of an outer surface of the upper frame **82** and an upper end of an outer surface of the lower frame **83** may be coupled to each other at the protruding and recessed portions thereof as if the lower frame **83** and the upper frame **82** are integrated with each other.

(294) Also, since the coupling end **831e** is disposed outside the coupling groove **831f**, the lifting assembly **84** may also be disposed at the corresponding position. Thus, when the upper frame **82** and the lower frame **83** are coupled to each other, the scissors fixing member **826** of the upper frame **82** and the scissors fixing member **836** of the lower frame **83** may not interfere with each other.

(295) As described above, the lower frame **83** and the upper frame **82** may have the same structure and shape except for the coupling groove **831f** and the coupling end **831e** and the disposed position of the scissors fixing member **836**.

(296) Also, the scissors fixing member **826** may be provided in an accommodation space **830** defined when the upper frame **82** and the lower frame **83** are coupled to each other. In detail, in a state in which the upper frame **82** completely moves downward, the outer edge **821a** of the upper frame **82** and the outer edge **821a** of the lower frame **83** may contact each other.

(297) Thus, the upper frame **82** and the lower frame **83** may contact each other to define the accommodation space **830** therein. The lifting assembly **84** may be accommodated in the

accommodation space **830** in a state of being completely folded. That is, in the state in which the elevation device **80** descends to the lowermost position, the rest of the constituents of the lifting assembly **84** except for the first sliding shaft **842** and the second sliding shaft **844** may be disposed in the space defined by the frame parts of the upper frame **82** and the lower frame **83**. Also, the first sliding shaft **842** and the second sliding shaft **846** may also be disposed in the spaces **823**, **824**, **833**, and **834** defined by the partition parts **822** and **832**.

(298) Thus, the additional space for accommodating the lifting assembly **84** in addition to the upper frame **82** and the lower frame **83** may not be required so that the loss of storage space inside the drawer part **32** is minimized.

(299) Furthermore, since the support plate **81** also has a structure capable of accommodating the upper frame **82** and/or the lower frame **83**, a space for arranging the upper frame **82** and the lower frame **83** may not be additionally required. Thus, a loss in space for disposing the elevation device **80** may be minimized to secure the storage capacity of the drawer part **32**.

(300) That is, even if the elevation device **80** having the complicated scissors type is disposed, a space loss equivalent to the thickness of the support plate **81** or a space occupied by the coupling between the upper frame **82** and the lower frame **83** may be generated to very effectively utilize the interior of the drawer part **32**.

(301) An elevation device fixing part **837** may be disposed on the bottom surface of the frame part **821** of the lower frame **83**. The elevation device fixing part **837** may have an opened hole shape and have a protruding shape protruding from the bottom surface of the drawer part **32** when the elevation device **80** is mounted inside the drawer part **32** and may be combined in shape with an elevation device coupling part (not shown). That is, the elevation device **80** may be fixed to match the inside of the drawer part **32** by a simple operation that is seated inside the drawer part **32** and be maintained in the stable state even though the elevation device **80** operates. Also, the elevation device **80** may be easily lifted and separated from the drawer part **32** without any additional tool even if the elevation device **80** is not disposed in the drawer part **32**.

(302) The scissors assemblies **84** may be provided on both left and right sides of the lifting assembly **84**. The scissors assemblies **84** may be connected to the connection assembly **70** and may be independently driven by the power transmitted through the shaft **41** and the lever **42** to lift the upper frame **82**. Here, the scissors assemblies **84** on both sides may not cause any misalignment or deviation in one of the driving motors **64** and the structure of the driving device **40** including the shaft **41** and the screw assembly **50** so as to provide a structure capable of being elevated by the same height.

(303) Thus, the pair of scissors assemblies **84** disposed on both sides may be exerted independently by the pair of the levers **42** operating at the same time even when supporting heavy loads, and thus, the pair of scissors assemblies **84** may be elevated at the same time. Here, the lifting assembly **84** may be configured to allow the upper frame **82**, i.e., the support plate **81** to be elevated in the horizontal state.

(304) FIG. **28** is a perspective view of the scissors assembly.

(305) Referring to FIG. **28**, the lifting assembly **84** may include a pair of first rods **841** arranged in parallel to each other, a first sliding shaft **842** connecting both ends of the first rod **841**, and a first rotation shaft **843**.

(306) Each of the first rod **841**, the first sliding shaft **842**, and the first rotation shaft **843** may have a width that is enough to be accommodated inside the frame part **821**. Also, the first rod **841** may be disposed in a region corresponding to that of the frame part **821**, and the first rotation shaft **843** may also be disposed at an region corresponding to the frame part **821**.

(307) Also, the rod protrusion **841a** and the scissors protrusion **841b** may be disposed on one end of the first rod **841**. Here, the rod protrusion **841a** may be disposed on the same extension line as the first rotation shaft **843**, and the first rotation shaft **843** may rotate when the rod protrusion **841a** rotates.

(308) The first rotation shaft **843** may further include a rotation enhancing part **843a**. The rotation enhancing part **843a** may be configured to connect a portion of the first rod **841** to the entire first rotation shaft **843**. Thus, when the first rod **841** rotates, the first rotation shaft **843** may rotate together and also be enhanced to withstand the generated moment.

(309) Also, a mounting hole **843b** may be defined in each of both ends of the rotation enhancing part **843a**, and the scissors fixing member **836** may be mounted to pass through the mounting hole **843b**. Thus, the first rotation shaft **843** may be rotatably mounted on the scissors fixing member **836** of the lower frame **83**.

(310) The first sliding shaft **842** may connect the other end of the first rod **841** and may be disposed to pass through the slide guide **825**. Thus, the first sliding shaft **842** may move along the slide guide **825** of the upper frame **82** when the first rod **841** rotates.

(311) Also, a first roller **841c** may be disposed on an upper end of the first rod **841** to which the first sliding shaft **842** is connected. The first roller **841c** may contact an inner surface of the frame part **821**. When the first sliding shaft **842** moves along the slide guide **825**, the first roller **841c** may contact the inner surface of the frame part **821** to rotate. Thus, the elevation device **80** may smoothly operate even though the upper frame **82** is pressed by a weight object.

(312) The first sliding shaft **842** may further include an elastic member mounting part **842a**. The elastic member mounting part **842a** may be configured to fix a scissors elastic member **85** connecting the inner surface of the frame part **821**, to which the first sliding shaft **842** and the second rod **844** are rotatably fixed. A mounting hole **842b** may be formed in the elastic member mounting part **842a** to fix one end of the scissors elastic member **85**.

(313) As illustrated in FIG. 23, since the frame part **821** and the first sliding shaft **842** are disposed at the farthest positions in a state in which the elevation device **80** is disposed at the lowest height, the scissors elastic member **85** may be in the maximumly tensioned state. Thus, when the upper frame **82** ascends, the restoring force of the scissors elastic member **85** may provide the additional force so that the elevation device **80** is elevated with less force.

(314) As illustrated in FIG. 24, the first sliding shaft **842** may be disposed closest to the frame part **821** in a state in which the elevation device **80** is disposed at the maximum height, and thus, the scissors elastic member **85** may be in the minimally tensioned state or the non-tensioned state. Since the scissors elastic member **85** is tensioned to allow the upper frame **82** to descend when the elevation device **80** descends, the upper frame **82** may slowly descend by the elastic force of the scissors elastic member **85**. Thus, even though the heavy food is accommodated, the elevation device **80** may be buffered to reduce impact noise and smoothly descend.

(315) Also, a pair of second rods **844** may be provided to cross the first rod **841**. The first rod **841** and the second rod **844** may be connected to each other by the scissors shaft **845** so that the first rod **841** and the second rod **844** rotate in the state of crossing each other. The second rod **844** may be disposed between both the first rods **841**. That is, the scissors shaft **845** may sequentially pass through the first rod **841** and the second rod **844** from the outside of the first rod **841**.

(316) A second sliding shaft **842** connecting an upper end of the second rod **844** and a second rotation shaft **847** connecting a lower end of the second rod **844** may be provided on the second rod **844**.

(317) The second rod **844**, the second sliding shaft **842**, and the second rotation shaft **847** may also have shapes and arrangements that are enough to be accommodated in the frame part **821**. In this state, both the second rotation shafts **847** connecting the upper ends of the second rods **844** may be provided.

(318) The second rotation shaft **847** may be rotatably mounted on the scissors fixing member **826** of the upper frame **82**. Here, the second rotation shaft **847** passing through the scissors fixing member **826** may further include a rotation bush **847a**. The rotation bush **847a** may contact the inner surface of the scissors fixing member **826** and may be made of a plastic material having excellent lubrication performance and abrasion resistance. Thus, the operation of the lifting

assembly **84** may be smoothly performed.

(319) The lower ends of the second rods **844** disposed on both sides may be connected by the second sliding shaft **842**. The second sliding shaft **846** may be mounted to pass through the slide guide **835** provided in the lower frame **83** and may move along the slide guide **835** as the elevation device **80** is elevated.

(320) Also, a second roller **844c** may be disposed on an upper end of the second rod **844** to which the second sliding shaft **846** is connected. The second roller **844c** may contact an inner surface of the frame part **831**. When the first sliding shaft **842** moves along the slide guide **835**, the first roller **841c** may contact the inner surface of the frame part **831** to rotate. Thus, the elevation device **80** may smoothly operate even though the lower frame **83** is pressed by a weight object.

(321) FIG. **29** is a perspective view illustrating a state in which the elevation device ascends when viewed from a lower side. Also, FIG. **30** is an enlarged view of a portion “B” of FIG. **29**. Also, FIG. **31** is an enlarged view of a portion “C” of FIG. **29**. Also, FIG. **32** is a partial perspective view of one side of a lower portion in the state in which the elevation device ascends.

(322) In details of the coupling structure of the lifting assembly **84**, the upper frame **82**, and the lower frame **83** with reference to the accompanying drawings, the lifting assembly **84** provided by coupling the first rod **841** to the second rod **844** may be disposed on both left and right sides between the upper frame **82** and the lower frame **83** and then be folded or unfolded at the same time to elevate the upper frame **82**.

(323) As illustrated in FIGS. **30** and **31**, an upper end of the second rod **844** may be disposed on each of both left and right ends of the upper frame **82**. Here, the upper end of the second rod **844** may be fixed and mounted in a state of being accommodated in the inner surface of the frame part **821**.

(324) Also, the second rotation shaft **847** connecting both sides of the upper end of the second rod **844** to each other may be mounted to pass through the pair of scissors fixing members **826** provided on the upper frame **82**. Here, a rotation bush **847a** may be provided between the scissors fixing member **826** and the upper frame **82** to allow the second rotation shaft **847** passing through the fixed fixing member **826** in the fixed state to rotate.

(325) Also, the second sliding shaft **846** connecting both sides of the lower end of the first rod **841** to each other may pass through the slide guide **835** and move along the slide guide **835** when the first rod **841** rotates. Here, the first roller **841c** on the upper end of the first rod **841** may contact the inner bottom surface of the upper frame **82**, and when the first rod **841** rotates, the first roller **841c** may be rolled in the state of contacting the inner surface of the upper frame **82**.

(326) A plate-shaped guide part **848** may be disposed on a portion at which the upper end of the first rod **841** is connected to the first sliding shaft **842**. The movement of the first rod **841** in the left and right directions may be restricted by the guide part **848**. That is, the first rod **841** may be spaced apart from the inner side surface of the bent inner surface of both side surfaces of the upper frame **82** by a width of the guide part **848**. The guide part **848** may have a width corresponding to that of the second rod **844**. Thus, when the first rod **841** and the second rod **844** are folded, the first rod **841** and the second rod **844** may be disposed in the inner space of the frame part **821** without interfering with each other.

(327) As illustrated in FIG. **32**, the lower end of the first rod **841** may be disposed on both sides of the lower frame **83** and be accommodated inside the frame part **831**. Also, the first rotation shaft **843** connecting the lower end of the first rod **841** may be fixed to pass through the scissors fixing members **836** disposed on both side of the lower frame **83**.

(328) Here, since the first rod **841** is disposed outside the second rod **844**, the scissors fixing member **836** disposed on the lower frame **83** may be disposed outside the scissors fixing member **826** disposed on the upper frame **82**. Thus, when the upper frame **82** descends, the scissors fixing members **826** and **836** may not interfere with each other. Also, the coupling ends **821e** and **831e** may also be disposed inside and outside with respect to each other by the positions of the scissors

fixing members **826** and **836** so as not to interfere with each other. Also, the coupling ends **821e** and **831e** and the coupling grooves **821f** and **831f** facing each other at the ends of the upper frame **82** and the lower frame **83** may have shapes corresponding to each other and may be coupled to each other when the upper frame **82** descends.

(329) The second roller **844c** may be provided on the lower end of the second rod **844** to maintain the contact state with the inner surface of the lower frame **834**. Also, when the second sliding shaft **846** moves along the slide guide **835**, the second roller **844c** may be rolled along the inner surface of the lower frame **83** while being accommodated inside the frame part **831**.

(330) As described above, the lower end of the first rod **841** and the upper end of the second rod **844** may be fixed to the lower frame **83** and the upper frame **82** in a rotatable manner. Also, when the elevation device **80** operates, the first rod **841** and the second rod **844** may rotate. Here, the first roller **841c** and the second roller **844c** may rotate while contacting the upper frame **82** and the lower frame **83** so that the first and second sliding shafts **842** and **846** smoothly rotate.

(331) The first rod **841** and the second rod **844** may be disposed inside the pair of the first rods **841**. When the rod **804** completely descends, all of the first rod **841** and the second rod **844** may be accommodated in an accommodation space **830** defined by the frame parts **821** and **831**.

(332) FIG. **33** is a cross-sectional view taken along line **33-33'** of FIG. **23**.

(333) Referring to FIGS. **32** and **33**, the first rod **841** may be disposed outside the second rod **844**. Also, all of the ends of the first rod **841** and the second rod **844** may be accommodated inside the frame part **831** of the lower frame **83**.

(334) In detail, the first rod **841** and the second rod **844** may be disposed in parallel to each other, and the sum of a width **W2** of the first rod **841** and a width **W3** of the second rod **844** may correspond to or slightly less than a width of the accommodation space **830**.

(335) Thus, while the elevation device **80** operates to allow the first rod **841** and the second rod **844** to contact each other, the first rod **841** and the second rod **844** may not interfere with each other and also be completely accommodated in the frame part **831**.

(336) Also, a height of the space of each of the frame parts **821** and **831** between the upper frame **82** and the lower frame **83** in the state in which the upper frame **82** and the lower frame **83** are coupled to each other may be greater than the sum of thicknesses of the first rod **841** and the second rod **844** or a height in the state in which the first rod **841** and the second rod **844** are completely folded.

(337) In addition, a length **L2** of each of the first rod **841** and the second rod **844** may be equal to or less than a length **L1** from an outer end to a central portion of each of the upper frame **82** and the lower frame **83** or a length from an outer end of each of the upper frame **82** and the lower frame **83** to each of the partition parts **822** and **832**.

(338) Due to this structure, in the state in which the upper frame **82** completely descends, the first rod **841** and the second rod **844** may be completely accommodated in the accommodation space **830** within the frame parts **821** and **831**, which is defined by coupling the upper frame **82** to the lower frame **83**. Here, the first rotation shaft **843** and the second rotation shaft **847** may be accommodated inside the frame parts **821** and **831**, which define the left and right sides of the upper frame **82** and the lower frame **83**.

(339) Hereinafter, the selective coupling and power connection of the elevation device **80** and the connection assembly **70** will be described in more detail with reference to the drawings.

(340) FIG. **34** is a perspective view illustrating a connection state between the connection assembly and the elevation device. Also, FIG. **35** is a cross-sectional view illustrating the connection state between the connection assembly and the elevation device. Also, FIG. **36** is a perspective view illustrating a separation state of the connection assembly and the elevation device.

(341) As illustrated in the drawings, if the service of the driving device **40** or the elevating device **80** is necessary or if the use of the elevation device **80** is not desired, the driving device **40** and the elevation device **80** may be simply separated from and coupled to each other.

(342) As illustrated in FIGS. **34** and **35**, the front panel door part **31** and the drawer part **32** may be coupled to each other, and power transmission may be possible in the state in which the connection assembly **70** and the elevation device **80** are connected to each other. Here, the connection member **73** may be connected to the lever **42** and the elevation device **80**, and the first connection part **731** may be connected to the fixing shaft **77** and the rod protrusion **841a** of the elevation device **80**. The lever protrusion **425** and the scissors protrusion **841b** may be inserted into the second connection part **732**.

(343) In this state, when the lever **42** rotates by the operation of the driving device **40**, the rod protrusion **841a** of the elevation device **80** may rotate by the first connection part **731**, and the lifting assembly **84** of the elevation device **80** may rotate.

(344) Here, since the second connection part **732** is connected to the scissors protrusion **841b** of the elevation device **80**, greater force may be transmitted to the elevation device **80**. In detail, the second connection part **732** may be disposed at a position away from the first connection part **731**, and thus when the first connection part **731** rotates around the shaft, a moment similar to a leverage may be applied to the second connection part **732**. Thus, a moment greater than the moment generated at the first connection part **731** may be applied together with the second connection part **732**, and thus the elevation device **80** may rotate with larger force.

(345) Furthermore, since the pair of scissors assemblies **84** are disposed on both sides of the lifting assembly **84**, the power may be transmitted to the lifting assembly **84**, thereby effectively elevating the elevation device **80** with less force.

(346) The connection member **73** may have a single shaft structure that connects the lever **42** to the rod protrusion **841a** of the elevation device **80** when the torque by the driving device **40** is sufficient. The lifting assembly **84** may also be configured so that the connection member **73** is connected to each of both sides of one of the scissors assemblies **84** to elevate the elevation device **80**.

(347) The user may push the push member **74** of the connection assembly **70** to push the connection member **73** in the state in which the service condition of the driving device or the elevating device **80** of the refrigerator **1** occurs. The coupling between the connection member **73** and the elevation device **80** may be released by allowing the connection member **73** to move forward.

(348) In this state, the front panel door part **31** may be separated from the drawer part **32**, and the entire driving part **40** provided in the front panel door part **31** may be completely separated from the drawer part **32** by a single operation.

(349) The driving part **40** may be maintained in the state in which the front panel door part **31** is separated, and the front panel door part **31**, which normally operates as necessary, may be replaced to be mounted. Here, the connection member **73** of the front panel door part **31** may be coupled to the rod protrusion **841a** and the scissors protrusion **841b** of the elevation device without separate assembly and disassembly.

(350) The front panel door part **31** and the drawer part **32** may be rigidly coupled to each other by the door frame or other structure, and the front panel door part **31** and the drawer part **32** may be additionally separated from or coupled to each other when the front panel door part **31** and the drawer part **32** are separated from or coupled to each other.

(351) Hereinafter, a state in which the door **30** of the refrigerator **1** is inserted and withdrawn and is elevated according to an embodiment will be described in more detail with reference to the accompanying drawings.

(352) FIG. **37** is a perspective view illustrating a state in which the drawer door is closed.

(353) As illustrated in the drawing, in the state in which the food is stored, the refrigerator **1** may be maintained in a state in which all of the rotation door **20** and the door **30** are closed. In this state, the user may withdraw the door **30** to accommodate the food.

(354) The door **30** may be provided in plurality in a vertical direction and be withdrawn to be

opened by the user's manipulation. Here, the user's manipulation may be performed by touching the manipulation part **301** disposed on the front surface of the rotation door **20** or the door **30**. Alternatively, an opening command may be inputted on the manipulation device **302** provided on the lower end of the door **30**. Also, the manipulation part **301** and the manipulation device **302** may individually manipulate the insertion and withdrawal of the door **30** and the elevation of the elevation member frame space **830**. Alternatively, the user may hold a handle of the door **30** to open the drawer door **30**.

(355) Hereinafter, although the lowermost door **30** of the doors **30**, which are disposed in the vertical direction, is opened and elevated as an example, all of the upper and lower doors **30** may be inserted and withdrawn and elevated in the same manner.

(356) FIG. **38** is a perspective view illustrating a state in which the drawer door is completely opened. Also, FIG. **39** is a cross-sectional view illustrating a state of the drawer door in a state in which the basket of the drawer door completely descends.

(357) As illustrated in the drawings, the user may manipulate the draw-out operation on the door **30** to withdraw the door **30** forward. The door **30** may be withdrawn while the draw-out rail **33** extends.

(358) The door **30** may be configured to be inserted and withdrawn by the driving of the draw-out motor **14**, not by a method of directly pulling the door **30** by the user. The draw-out rack **34** provided on the bottom surface of the door **30** may be coupled to the pinion gear **141** rotating when the draw-out motor **14** provided in the cabinet **10** is driven. Thus, the door **30** may be inserted and withdrawn according to the driving of the draw-out motor **14**.

(359) The draw-out distance of the door **30** may correspond to a distance at which the front space **S1** within the door **30** is completely exposed to the outside. Thus, in this state, when the elevation device **80** is elevated, the container or the food may not interfere with the doors **20** and **30** or the cabinet **10** disposed thereabove.

(360) Here, draw-out distance of the door **30** may be determined by a draw-out detection device **15** disposed on the cabinet **10** and/or the door **30**. The draw-out detection device **15** may be provided as a detection sensor that detects a magnet **389** to detect a state in which the door **30** is completely withdrawn or closed.

(361) For example, as illustrated in the drawings, the magnet **389** may be disposed on the bottom of the drawer part **32**, and the detection sensor may be disposed on the cabinet **10**. The draw-out detection device **15** may be disposed at a position corresponding to a position of the magnet **389** when the door **30** is closed and a position of the magnet **389** when the door **30** is completely withdrawn. Thus, the draw-out state of the door **30** may be determined by the draw-out detection device **15**.

(362) Also, as necessary, a switch may be provided at each of positions at which the door **30** is completely inserted and withdrawn detect the draw-out state of the door **30**. In addition, the draw-out state of the door **30** may be detected by counting the rotation number of draw-out motor **14** or measuring a distance between the rear surface of the front panel door part **31** and the front end of the cabinet **10**.

(363) In the state in which the door **30** is completely withdrawn, the elevation motor **64** may be driven to elevate the elevation device **80**. The elevation device **80** may be driven in an even situation in which the door **30** is sufficiently withdrawn to secure safe elevation of the food or container **36** seated on the elevation device **80**.

(364) That is, in the state in which the door **30** is withdrawn to completely expose the front space **S1** to the outside, the elevation device **80** may ascend to prevent the container **36** or the stored food seated on the elevation device **80** from interfering with the doors **20** and **30** or the cabinet **10**.

(365) In details of the draw-out state of the door **30**, the front space **S1** has to be completely withdrawn to the outside of the lower storage space **12** in the state in which the door **30** is withdrawn for the elevation.

(366) Particularly, the rear end L1 of the front space S1 has to be more withdrawn than the front end L2 of the cabinet 10 or the upper door 20. Also, the rear end L1 of the front space S1 is disposed at a further front side than the front end L2 of the cabinet 10 or the door 20 so as to prevent the elevation device 80 from interfering when the elevation device 80 is elevated.

(367) Also, when the elevating device 80 is completely withdrawn to be driven, the entire drawer part 32 may not be completely withdrawn but withdrawn up to only a position for avoiding interference when the elevating device 80 is elevated as illustrated in FIG. 39. Here, at least a portion of the rear space S2 of the drawer part 32 may be disposed inside the lower storage space 12. That is, the rear end L3 of the drawer part 32 may be disposed at least inside the lower storage space 12.

(368) Thus, even when the weight of the stored object is added to the weight of the door 30 itself including the driving device 40 and the elevation device 80, the deflection or damage of the draw-out rail 33 or the door 30 itself may not occur to secure the reliable draw-out operation.

(369) The ascending of the elevation device 80 may start in a state in which the door 30 is completely withdrawn. Also, to secure the user's safety and prevent the food from being damaged, the ascending of the elevation device 80 may start after a set time elapses after the door 30 is completely withdrawn.

(370) After the door 30 is completely withdrawn, the user may manipulate the manipulation part 301 to input the ascending of the elevation device 80. That is, the manipulation part 301 may be manipulated to withdraw the door 30, and the manipulation part 301 may be manipulated again to elevate the elevation device 80.

(371) Also, in the state in which the door 30 is manually inserted and withdrawn, the manipulation part 301 may be manipulated to elevate the elevation device 80.

(372) As illustrated in FIG. 39, the driving device 40 and the elevation device 80 may not operate until the door 30 is completely withdrawn, and the elevation device 80 may be maintained in the lowest state.

(373) FIG. 40 is a perspective view illustrating a state of the driving device in the state in which the basket of the drawer door completely descends. Also, FIG. 41 is a perspective view illustrating a state of the driving device in the state in which the basket of the drawer door completely descends.

(374) As illustrated in the drawings, when a signal for operating the driving device 40 is not input, the driving device 40 may not operate to maintain the elevation device 80 at the lowest state.

(375) In this state, the driving device 40 may not operate, and the screw holder 56 may be disposed at the lowest position of the screw 52 as illustrated in FIG. 32. Also, the lever 42 may not also rotate, and the first connection part 731 and the second connection part 732 of the connecting member 73 may be disposed on the same height.

(376) The elevation device 80 may maintain the current state as long as the upper frame 82 is in the lowest state as shown in FIG. 41, and the driving device 40 may not operate.

(377) In this state, the upper frame 82 and the lower frame 83 may contact each other, and the lifting assembly 84 may be accommodated in the upper frame 82 and the lower frame 83.

(378) Here, no external force may be applied to the rod protrusion 841a and the scissors protrusion 841b of the elevation device 80 connected to the connection assembly 70 because the lever 42 does not operate.

(379) The end of the frame part 821 and the sliding shaft 842 may be farthest away from each other, and the scissors elastic member 85 may be in the maximum tensioned state. Thus, the upper frame 82 may be elevated more effectively by the restoring force of the scissors elastic member 85 in addition to the power transmission by the driving device 40 when the driving device 40 is started to be driven.

(380) A signal input for the elevation by the user may be waited while maintaining the above-described state, and the driving device 40 may operate when the user input the elevation manipulation.

(381) FIG. 42 is a cross-sectional view illustrating a state of the drawer door in a state in which the basket of the drawer door completely ascends.

(382) As illustrated in FIG. 39, in the state in which the door 30 is withdrawn, when the operation signal of the driving device is inputted, the driving device 40 may operate, and the state as illustrated in FIG. 42 may be obtained by elevating the elevation device 80.

(383) The driving device 40 may be connected to the elevation device 80 by the connection assembly 70 so that the power is transmitted to the elevation device 80. The power may be transmitted to the elevation device 80 by the connection assembly 70 together with the operation of the driving device 40, and the elevation device 80 may start to ascend.

(384) The elevation device 80 may continuously ascend and then be stopped when ascend to a sufficient height to facilitate access to the food or container 36 seated on the elevation device 80 as illustrated in FIG. 34. In this state, the user may easily lift the food or container 36 without overtaxing the waist.

(385) When the elevation completion signal of the elevation device 80 is inputted, the driving of the driving motor 64 may be stopped. For this, a height detection device 16 capable of detecting the position of the elevation device 80 may be provided. The height detection device 16 may be provided on the front panel door part 31 and may be disposed at a position corresponding to the maximum height of the elevation device 80 and at a position corresponding to the lowest height of the elevation device 80.

(386) The height detection device 16 may be provided as a detection sensor that detects a magnet 389. The height detection device 16 may detect the magnet 389 disposed on the elevation device 80 to determine whether the ascending of the elevation device 80 is completed. Also, the height detection device 16 may be provided as a switch structure to turn on the switch when the elevation device 80 maximally ascends. Also, the height detection device 16 may be provided on the elevation rail 44 or the screw 52 to detect the maximally ascending position of the elevation device 80. Also, whether the elevation device 80 maximally ascends may be determined according to a variation in load applied to the elevation motor 64.

(387) The driving of the elevation motor 64 is stopped in the state in which the elevation device 80 maximally ascends. In this state, although the elevation device 80 is disposed inside the drawer part 32, the food or container 36 seated on the elevation device 80 may be disposed at a position higher than the opened top surface of the drawer part 32. Thus, the user may easily access the food or container 36. Particularly, it is not necessary to allow the waist excessively for lifting the container 36, so that it is possible to perform safer and more convenient operation.

(388) In details of the maximally ascending state of the elevation device 80, the elevation device 80 may be elevated by driving the driving device 40 and be disposed at least at a lower position than the upper end of the drawer part 32.

(389) In the elevation device 80, when viewed with respect to the container 36 in the state in which the container 36 is seated, the upper end H1 of the container 36 may ascend to a position higher than the upper end H2 of the lower storage space 12. Here, the height of the container 36 may reach a height suitable for the user to reach the container 36 without stretching his/her waist.

(390) That is, the driving device 40 may have a structure in which the container 36 ascends from the inside of the drawer part 32. However, when the container 36 is mounted on the elevation device 80, the container 36 may be disposed at an accessible height.

(391) After the user's food storing operation is completed, the user may allow the elevation device 80 to descend by manipulating the manipulation part 301. The descending of the elevation device 80 may be performed by reverse rotation of the elevation motor 64 and may be gradually performed through the reverse procedure with respect to the above-described procedure.

(392) Also, when the descending of the elevation device 80 is completed, i.e., in the state of FIG. 39, the completion of the descending of the elevation device 80 may be performed by the height detection device 16. The height detection device 16 may be further provided at a position that

detects the magnet disposed on the elevation device **80** when the elevation device **80** is disposed at the lowermost descending position. Thus, when the completion of the descending of the elevation device **80** is detected, the driving of the driving device **40** is stopped.

(393) Also, after the driving of the elevation motor **64** is stopped, the door **30** may be inserted. Here, the door **30** may be closed by the user's manipulation or by the driving of the draw-out motor **14**. When the door **30** is completely closed, a state of FIG. **29** may become.

(394) Hereinafter, a state in which the elevation device **80** is elevated by the operation of the driving device will be described.

(395) FIG. **43** is a perspective view illustrating a state of the driving device in the state in which the basket of the drawer door completely ascends. Also, FIG. **44** is a perspective view illustrating a state of the elevation device in the state in which the basket of the drawer door completely ascends.

(396) As illustrated in the drawings, when the driving motor **64** rotates in the normal or reverse direction by the ascending/descending signal of the elevation device **80**, the operation of the driving device **40** may start.

(397) In details of the ascending operation of the elevation device **80**, the shaft **41** may rotate by the operation of the driving motor **64**. The shaft gear **412** on both sides of the shaft **41** and the screw gear **53** may rotate in the state of being engaged with each other by the rotation of the shaft **41**, the screw **52** may rotate.

(398) Here, both the shaft gears **412** may be connected to both ends of the shaft **41** to rotate. Thus, the same rotation force may be transmitted to the both the screw assemblies **50**. Also, in the screw assembly **50** having the same structure, the screw **52** may rotate at the same rotational speed, and the screw holder **56** may ascend by the same height at the same time.

(399) As the screw holder **56** ascends, the lever **42** connected to the screw holder **56** may also rotate. The lever **42** may continuously rotate in the state of being axially coupled to the connection assembly **70**, and the connection member **73** connected to the lever **42** may also rotate together with the rotation of the lever **42**.

(400) The connection member **73** may be connected to the lever protrusion **425** at a position apart from the rotation shaft of the lever **42** as well as the rotation shaft of the lever **42** by the lever protrusion **425**. Thus, the elevation device **80** connected to the connection member **73** and the connecting member **73** may rotate with a larger moment.

(401) The rod protrusion **841a** of the elevation device **80** and the scissors protrusion **841b** may be coupled to the first connection part **731** and the second connection part **732** of the connection member **73**. The rotational force may be transmitted to the first rod **841** of the lifting assembly **84** by rotation of the lifting assembly **84** so that the lifting assembly **84** is more effectively unfolded.

(402) As the lifting assembly **84** is unfolded, the sliding shaft **842** may move toward the frame part **821** at a position adjacent to the side of the partition part **822**, and the first rod **841** and the second rod **844** may rotate in the direction in which an angle gradually increases.

(403) The sliding shaft **842** may be connected to the frame part **821** at the facing position by the scissors elastic member **85** and the restoring force of the scissors elastic member **85** may allow the sliding shaft **842** to move toward the frame part **821**. Thus, it is possible to assist the elevation of the elevating device **80**.

(404) As a result, the lifting assembly **84** may be unfolded so that the upper frame **82** may ascend, and the container **36** or the food placed on the support plate **81** may ascend and finally the elevation device **80** may ascend to its maximum height as illustrated in FIG. **44**.

(405) In this state, when the driving device **40** is stopped, and the user allows the elevating device **80** to move downward after storing the food, the driving motor **64** may rotate in the reverse direction and operate in the reverse order, and thus, the elevation device **80** may descend and then be in the state as illustrated in FIG. **39**.

(406) In addition to the foregoing embodiment, various embodiments may be exemplified.

(407) Hereinafter, another embodiments will be described with reference to the accompanying

drawings. In the other embodiments of the present disclosure, the same reference numerals are used for the same components as those of the above-described embodiments, and a detailed description thereof will be omitted.

(408) FIG. 45 is an exploded perspective view illustrating a coupling structure of an elevation device and a support plate according to a second embodiment. Also, FIG. 46 is an exploded perspective view of the elevation device. Also, FIG. 47 is a view illustrating an arrangement of a support plate in a state in which the elevation device descends at the lowest position.

(409) As illustrated in the drawings, an elevation device 80 according to a second embodiment may include an upper frame 82, a lower frame 83, and a pair of scissors assemblies 84. Also, the elevation device 80 may include a support plate 86 seated on the upper frame 82.

(410) A coupling structure of the upper frame 82, the lower frame 83, and the lifting assembly 84 may be the same as that according to the first embodiment, and the lower frame 83 may be fixed to the inside of the drawer part 32, and the upper frame 82 may be elevated according to the operation of the lifting assembly 84.

(411) The scissors assemblies 84 having the same structure may be disposed on both left and right sides, respectively. Also, one of the scissors assemblies 84 may include a pair of first rods 841 and a pair of second rods 844, and the first rod 841 and the second rod 844 may be axially coupled to each other to rotate.

(412) Also, the ends of the first rods 841 disposed on both sides may be connected by a first rotation shaft 843, and the ends of the second rods 844 disposed on both sides may be connected by a second rotation shaft 847. Also, the first rotation shaft 843 and the second rotation shaft 847 may be configured to pass through the scissors fixing members 826 and 836 mounted on the lower frame 83 and the upper frame 82, respectively. Thus, each of the first rod 841 and the second rod 844 may be rotatably disposed about one end of each of the upper frame 82 and the lower frame 83.

(413) Also, a first roller 841c and a second roller 844c may be provided at the other end of the first rod 841 and the second rod 844, respectively. The first roller 841c and the second roller 844c may contact the inner surface of the upper frame 82 and the inner surface of the lower frame 83. When the lifting assembly 84 operates, the first roller 841c and the second roller 844c may be rolled along the inner surfaces of the upper frame 82 and the lower frame 83, respectively.

(414) The first rod 841 and the second rod 844 and the first rotation shaft 843 and the second rotation shaft 847 may be accommodated in inner spaces of the frame parts 821 and 831 in which the upper frame 82 and the lower frame 83 are disposed. Thus, when the upper frame 82 completely descends, the lower end of the upper frame 82 and the upper end of the lower frame 83 may contact each other to define a space so that the lifting assembly 84 is accommodated in the frame spaces 830 of the frame parts 821 and 831.

(415) Thus, when the upper frame 82 and the lower frame 83 are coupled to each other, the lifting assembly 84 may be accommodated inside the frame parts 821 and 831 and may not be exposed to the outside. Thus, only the scissors protrusion 841b and the rod protrusion 841a to be coupled may be exposed to the outside.

(416) Also, in the state in which the upper frame 82 moves to the lowermost position so that the upper frame 82 and the lower frame 83 are coupled to each other, as illustrated in FIG. 45, the upper and lower ends of the upper frame 82 and the lower frame 83 may contact each other, and the accommodation space 830 may be defined.

(417) The support plate 86 may be seated on the upper frame 82. Here, the support surface 861 of the support plate 86 may have a shape corresponding to the inside of the frame space 830 and be inserted into the accommodation space 830.

(418) Also, an edge 862 of the support plate 86 may have a width corresponding to that of the upper surface of the frame part 821 of the upper frame 82. Thus, the edge 862 of the support plate 86 may be seated on the upper frame 82 so as to surround a circumference of the upper frame 82 from an upper side. As illustrated in FIG. 47, the support surface 811 may extend to the inner

bottom of the frame space **830**, i.e., the bottom surface of the lower frame **83**. Thus, only the space in which the upper frame **82** and the lower frame **83** are disposed may be lost, and the remaining space may be used as a storage space as a whole even when the elevation device **80** is mounted. (419) As described above, when compared with the above-described elevation device **80**, the elevation device **80** according to the second embodiment may have a structure which the slide guides **825** and **835** and the partition parts **822** and **832** are not provided, and also, the first sliding shaft **842** and the second sliding shaft **846** are omitted in the lifting assembly **84**.

(420) Also, the unexplained reference numerals may have the same configuration as the above-described embodiment.

(421) FIG. **48** is a view illustrating an operation of an elevation device of a refrigerator according to a third embodiment.

(422) As illustrated in the drawing, a refrigerator **1** according to a third embodiment may include a front panel door part **31** in which the driving device **40** is provided and an elevation device support part **320** extending backward from the front panel door part **31** to support the elevation device **80**.

(423) The elevation device support part **320** may have at least one surface capable of supporting the bottom surface of the elevation device **80**. Also, the elevation device support part **320** may be connected to the rear surface of the front panel door part **31** and have a plate shape or a frame shape extending backward so as to support the elevation device **80** from the lower side. Here, the elevation device **80** may be configured to be exposed to the outside when the door **30** is withdrawn while being fixed to the top surface of the elevation device support part **320**.

(424) Alternatively, the elevation device support part **320** may not have a simple plate-like structure but have a structure that covers the circumference of the elevation device **80** so that the elevation device **80** is not exposed.

(425) Also, the elevation device support part **320** may further include a draw-out rail and a draw-out rack **34**. The door **30** may be inserted and withdrawn by a draw-out motor **14** and a pinion gear **141** provided in the cabinet **10**.

(426) The elevation device **80** may be fixedly mounted on the elevation device support part **320** and may be inserted and withdrawn when the door **30** is inserted and withdrawn. The elevation device **80** may have the same structure as that of the elevation device according to the foregoing embodiment. The elevation device **80** may be connected to the connection assembly **70** so as to be elevated according to the operation of the driving device **40**. Also, a support plate **81** may be provided on the top surface of the upper frame **82**, and the container **36** may be mounted on the support plate **81** to be elevated.

(427) That is, according to the third embodiment, unlike the drawer part **32** according to the foregoing embodiment, an elevation device support part **320** supporting the elevation device **80** may be provided.

(428) FIG. **49** is a perspective view of a refrigerator according to a fourth embodiment.

(429) As illustrated in the drawing, a refrigerator **1** according to a fourth embodiment may include a cabinet **10** having a storage space that is vertically partitioned and a door **2** opening and closing the storage space.

(430) The door **2** may include a rotation door **20** which is provided in an upper portion of a front surface of the cabinet **10** to open and close an upper storage space and a door **30** disposed in a lower portion of the front surface of the cabinet **10** to open and close a lower storage space. The door **30** may be inserted and withdrawn forward and backward in the above embodiment, and the container and the food inside the drawer part **32** may be vertically elevated by the operation of the driving device **40** and the elevation device **80** inside the door **30**.

(431) The elevation device **80** may be provided in the region of the front space of the inside of the drawer part **32**. Thus, the elevation device **80** may elevate the food in the region of the front space among the entire region of the drawer part **32**.

(432) A manipulation part **301** or a manipulation device **302** may be provided at one side of the

front panel door part **31**, and the driving part **40** may be installed inside the front panel door part **31**. Also, the pulling-out operation of the drawer door **30** and/or the elevation of the elevation device **80** may be carried out by the manipulation of the manipulation part **301** or the manipulation device **302**.

(433) The drawer part **32** may be provided with the elevation device **80**. The elevation device **80** may be elevated by a connection assembly that connects the driving device to the elevation device. Since the constituent of the drawer door **30** and constituent of the driving device **40** and the elevation device **80** are the same as those according to the foregoing embodiment, their detailed descriptions will be omitted.

(434) A plurality of containers **361** may be provided in the elevation device **80**. The container **361** may be a sealed container such as a kimchi passage, and a plurality of the containers **361** may be seated on the elevation device **80**. The container **361** may be elevated together with the elevation device **80** when the elevation device **35** is elevated. Thus, in the state in which the container **361** ascends, at least a portion of the drawer part **32** may protrude, and thus, the user may easily lift the container **361**.

(435) The elevation device **80** may interfere with the rotation door **20** in the rotation door **20** is opened even though the drawer door **30** is withdrawn. Thus, the elevation device **80** may ascend in a state in which the rotation door **20** is closed. For this, a door switch for detecting the opening/closing of the rotation door **20** may be further provided.

(436) FIG. **50** is a perspective view of a refrigerator according to a fifth embodiment.

(437) As illustrated in the drawings, a refrigerator **1** according to a fifth embodiment includes a cabinet **10** defining a storage space therein and a door **2** opening and closing an opened front surface of the cabinet **10**, which define an outer appearance of the refrigerator **1**.

(438) The door **2** may include a drawer door **30** that defines an entire outer appearance of the refrigerator **1** in a state in which the door **2** is closed and is withdrawn forward and backward. A plurality of the drawer doors **30** may be continuously arranged in the vertical direction. Also, the drawer doors **30** may be independently withdrawn by the user's manipulation. The drawer door **30** may be provided with the driving device **40** and the elevation device **80**.

(439) The driving part **40** may be installed in the front panel door part **31**, and the elevation part **80** may be provided inside the drawer part **32**. Also, the driving device **40** and the elevation device **80** may be connected to each other by the connection assembly **70** when the front panel door part **31** and the drawer part **32** are coupled to each other. Also, the elevation device **80** may be disposed in the front space **S1** of the total storage space of the drawer part **32**.

(440) The insertion and withdrawal of the drawer door **30** and the elevation of the elevation device **80** may be individually performed. After the drawer door **30** is withdrawn, the elevation device **80** may ascend. Then, after the elevation device **80** descends, the insertion of the drawer door **30** may be continuously performed.

(441) Also, when the plurality of drawer doors **30** are vertically arranged, the elevation device **80** inside the drawer door **30**, which is relatively downwardly disposed, may be prevented from ascending in a state where the drawer door **30** is relatively drawn upward. Thus, the drawer door **30** may be prevented from interfering with the drawer door **30** in which the food and container are withdrawn upward.

(442) Also, although the elevation device **80** ascends in the state in which the drawer door **30** that is disposed at the uppermost side is withdrawn in FIG. **38**, all of the drawer doors **30** disposed at the upper side may also be elevated by the elevation device **80** that is provided inside.

(443) If a height of each of the drawer doors **30** disposed at the upper side is sufficiently high, only the drawer door **30** disposed at the lowermost position or the elevation device **35** of the of drawer doors **30** disposed relatively downward may be elevated.

(444) FIG. **51** is a perspective view of a refrigerator according to a sixth embodiment.

(445) As illustrated in the drawings, a refrigerator **1** according to a sixth embodiment includes a

cabinet **10** defining a storage space therein and a door **2** opening and closing an opened front surface of the cabinet **10**, which define an outer appearance of the refrigerator **1**.

(446) The inside of the cabinet **10** may be divided into an upper space and a lower space. If necessary, the upper and lower storage spaces may be divided again into left and right spaces.

(447) The door **2** may include a rotation door **20** which is provided in an upper portion of the cabinet **10** to open and close the upper storage space and a drawer door **2** disposed in a lower portion of the cabinet **10** to open and close the lower storage space.

(448) Also, the lower space of the cabinet may be divided into left and right spaces. The drawer door **30** may be provided in a pair so that the pair of drawer doors **30** respectively open and close the lower spaces. A pair of the drawer doors **30** may be arranged on both sides of the right and left sides of the drawer door **30**. The drawer door **30** may include the driving device **40** and an elevation device **80**.

(449) The driving part **40** may be installed in the front panel door part **31**, and the elevation part **80** may be provided inside the drawer part **32**. Also, the driving device **40** and the elevation device **80** may be connected to each other by the connection assembly **70** when the front panel door part **31** and the drawer part **32** are coupled to each other. Also, the elevation device **80** may be disposed in the front space **S1** of the total storage space of the drawer part **32**.

(450) The drawer door **30** may have the same structure as the drawer door according to the foregoing embodiment. Thus, the drawer door **30** may be inserted and withdrawn by user's manipulation. In the drawer door **30** is withdrawn, the elevation device **80** may ascend so that a user more easily accesses a food or container within the drawer door **30**.

(451) The refrigerator and the elevation device for the refrigerator according to the embodiment may have the following effects.

(452) The refrigerator according to the embodiment, the portion of the storage space within the drawer door may be elevated in the state in which the drawer door is withdrawn. Thus, when the food is accommodated in the drawer door disposed at the lower side, the user may not excessively turn its back to improve the convenience in use.

(453) Particularly, in order to lift the heavy-weight food or the container containing the food, the user has to lift the food or container with a lot of power. However, the elevation within the drawer door may ascend up to a convenient position by driving the driving device to prevent the user from being injured and significantly improve the convenience in use.

(454) Also, the driving device constituted by the electric devices for providing the power may be provided inside the door part, and the elevation device for the elevation may be provided inside the drawer part so that the driving device and the elevation device are not exposed to the outside to improve the outer appearance.

(455) Particularly, the driving device constituted by the electric devices may be disposed inside the door part, and it may be possible to prevent the user from accessing the door to prevent the occurrence of the safety accident.

(456) Also, the driving device may be provided in the door to block the noise and reduce noise during the use.

(457) Also, the driving part that occupies a large portion of the entire constituents may be disposed in the door part to minimize the storage capacity loss of the drawer part.

(458) Also, the elevation device may also have the structure that is compactly folded and accommodated in the frame in the state of descending to maximally secure the capacity of the accommodation part in which the food is accommodated. Also, the support plate may be configured to be inserted into the opened central portion of the frame. Thus, all the rest portions except for the circumferential portion of the frame may be substantially provided in the space for accommodating the food to maximize the storage capacity of the accommodation part.

(459) Also, the scissors assembly constituting the elevation device may have the structure in which the rod itself rotates by receiving the power of the driving device. Thus, the power may be used for

the rotation of the rod without a loss so that the scissors assembly efficiently operates and also is stably elevated even though the heavy food is accommodated therein.

(460) Also, the connection member for transmitting the power for elevating the elevation device may be transmitted from the first connection part and the second connection part to the rotation shaft and also from one side away from the rotation shaft to the rotation shaft. Thus, the force applied at the point away from the rotation shaft may be applied as greater force by the moment. Thus, there is an advantage that when the elevation device is elevated, the greater force is provided to realize the more easy and effective elevation operation.

(461) Also, the elevation device may be provided at a portion of the front portion of the drawer part. Thus, the elevation device may be elevated without interfering with the upper door or cabinet at the time of the elevation without being withdrawn to the extent that the drawer part is completely exposed to the outside. Also, it may be possible to prevent the deflection and durability limitation caused by excessive withdrawal of the heavy drawer door and also to prevent the loss of the cold air caused by excessive withdrawal of the drawer.

(462) In addition, since the entire drawer is not elected but a portion of the drawer is elevated, the entire elevation structure may be compact, and the lightweight structure may be used to minimize the loss of the storage capacity and maintain the simple structure.

(463) Also, the elevation device may be constituted by the pair of scissors assemblies disposed on both left and right sides, and the pair of scissors may operate at the same time by the driving device to perform the elevation operation in the state in which the food supported by the elevation device is stable without being eccentric or tilted to one side.

(464) Also, the slide guide for guiding the slidable movement of the rod when the elevation device operates may be provided on the frame, and the roller rolled along the inside of the frame may be provided on the end of the rod. Thus, the scissors assembly may very smoothly operate.

(465) Also, the scissors assembly may operate in the state of being accommodated in the accommodation part, and the top surface of the lifting assembly may be covered by the support plate to fundamentally prevent the lifting assembly from being exposed, thereby improving the outer appearance and securing the user's safety.

(466) Although embodiments have been described with reference to a number of illustrative embodiments thereof, it should be understood that numerous other modifications and embodiments can be devised by those skilled in the art that will fall within the spirit and scope of the principles of this disclosure. More particularly, various variations and modifications are possible in the component parts and/or arrangements of the subject combination arrangement within the scope of the disclosure, the drawings and the appended claims. In addition to variations and modifications in the component parts and/or arrangements, alternative uses will also be apparent to those skilled in the art.

Claims

1. A refrigerator comprising: a cabinet that defines a storage space; a drawer door including: a drawer part with an accommodation space for configured to insert into and withdraw from the storage space in a front-rear direction, and a front panel door coupled to a front portion of the drawer part and configured to open and close at least a portion of the storage space; an elevation device provided in the drawer part and configured to elevate an object stored in the accommodating space relative to a bottom surface of the drawer part; and a driving device located in the front panel door and configured to supply driving force to move the elevation device, wherein the elevation device comprises a plurality of rods disposed in the accommodation space, the plurality of rods comprising a first rod and a second rod rotatably coupled to and cross each other in a left-right direction, wherein the first rod is one of a pair of first rods that are spaced apart from each other in the front-rear direction and parallel to each other, wherein the second rod is one of a pair of second

rods that are spaced apart from each other in the front-rear direction and parallel to each other, wherein the elevation device further comprises: a first rotation shaft disposed between the pair of the first rods in the front-rear direction and rotatably connected to one end of the first rods at a fixed position in the drawer part, and a first sliding shaft disposed between the pair of the first rods in the front-rear direction, connected to the other end of the first rods and slidably moving in the left-right direction based on rotating of the first rotation shaft, and wherein the driving device comprises: a motor assembly disposed in the front panel door to generate a driving force to drive the elevation device, and a fixing shaft disposed at a position corresponding to the first rotation shaft in the front panel door in the front-rear direction and rotatably connected to the motor assembly and the first rotation shaft.

2. The refrigerator according to claim 1, wherein the first rod further has a rod protrusion extends in the front-rear direction so as to protrude from the one end of the first rod toward the front panel door.

3. The refrigerator according to claim 2, further comprising a connection assembly comprising a first connection part that is rotatably coupled to the fixing shaft and the rod protrusion so that the first rotation shaft is rotated and the first rod is rotated about the first rotation shaft.

4. The refrigerator according to claim 3, wherein the first connection part comprises: a first hollow part in one end of the first connection part into which the fixing shaft is inserted; and a second hollow part in the other end of the first connection part into which the rod protrusion is inserted.

5. The refrigerator according to claim 3, wherein the rod protrusion is disposed on a same extension line with the first rotation shaft.

6. The refrigerator according to claim 5, wherein the elevating device further comprises: a lower frame configured to be fixed to a bottom of the accommodation space of the drawer part, and an upper frame disposed vertically above the lower frame and configured to support a container for storing the object, wherein: the plurality of the rods is disposed between the upper frame and the lower frame, the one end of the first rod is connected to the first rotation shaft, the first rotation shaft is rotatably fixed to a portion of the lower frame, and the other end of the first rod and the first sliding shaft are slidably coupled to the upper frame.

7. The refrigerator according to claim 6, wherein the upper frame and the lower frame respectively comprise: an upper frame part and a lower frame part extending in the left-right direction; and an upper edge wall part protruding to form an inner space along an edge of the upper frame part; and a lower edge wall part protruding to form an inner space along an edge of the lower frame part, wherein the one end and the other end of the first rod and the one end and the other end of the second rod are disposed in the inner space defined by the upper edge wall part and the lower edge wall part respectively.

8. The refrigerator according to claim 7, wherein the first rod further comprises a scissors protrusion that is disposed on the one end of the first rod and protrudes toward the front panel door at a position spaced apart from the rod protrusion.

9. The refrigerator according to claim 8, wherein the connection assembly further comprising a second connection part spaced apart from the first connection part and coupled to the scissors protrusion.

10. The refrigerator according to claim 8, wherein the lower edge wall part comprises a first lower edge groove formed on an outer edge, and wherein the rod protrusion is configured to pass through the first lower edge groove.

11. The refrigerator according to claim 10, wherein the lower edge wall part further comprises a second lower edge groove formed on the outer edge at a position corresponding to the scissors protrusion, wherein the second lower edge groove is configured to be spaced apart from the first lower edge groove based on the first rod and the second rod being folded, and wherein the scissors protrusion is configured to pass through the second lower edge groove.

12. The refrigerator according to claim 11, wherein the upper edge wall part comprises an upper

edge groove formed at a position corresponding to the second lower edge groove.

13. The refrigerator according to claim 7, wherein a height between the upper frame and the lower frame in a state in which the upper frame and the lower frame are coupled to each other is greater than a sum of thickness of the first rod and the second rod or a height in a state in which the first rod and the second rod are folded.

14. The refrigerator according to claim 7, wherein the upper frame part and the lower frame part have a portion extending in the front-rear direction respectively, and wherein the upper edge wall part and the lower edge wall part further protrude along each edge of the upper and lower frame parts in the front-rear direction.

15. The refrigerator according to claim 14, wherein the elevation device further comprises a lower scissors fixing member disposed inside the inner space formed by the lower edge wall part in the front-rear direction, wherein the lower scissors fixing member is fixed to a portion of the lower edge wall part, and wherein the first rotation shaft passes through the lower scissors fixing member to be rotatably mounted in the lower frame.

16. The refrigerator according to claim 15, wherein a rotation enhancing part is formed to connect between the first rod and the first rotation shaft, and wherein the rotation enhancing part comprises a mounting hole formed in each of both end portions thereof.

17. The refrigerator according to claim 16, wherein the lower scissors fixing member passes through the mounting hole so that the first rotation shaft and the rotation enhancing part rotate in the lower frame.

18. The refrigerator according to claim 15, wherein the elevation device further comprises an upper scissors fixing member disposed inside the inner space formed by the upper edge wall part in the front-rear direction, wherein the upper scissors fixing member is fixed to a portion of the upper edge wall part, and wherein the upper scissors fixing member is located at a position not to interfere with the lower scissors fixing member in a state which the plurality of the rods are folded.

19. The refrigerator according to claim 15, wherein the lower edge wall part comprises a coupling end protruding upward from a top portion of the lower edge wall part, wherein the lower scissors fixing member is coupled to the coupling end, and wherein a coupling groove recessed from the top portion of the lower edge wall part is disposed a rear position of the coupling end in the front-rear direction.

20. A refrigerator comprising: a cabinet that defines a storage space; a drawer door comprising (i) a drawer part with an accommodation space configured to insert into and withdraw from the storage space in a front-rear direction, and (ii) a front panel door coupled to a front portion of the drawer part and configured to open and close at least a portion of the storage space; an elevation device provided in the drawer part and configured to elevate an object stored in the drawer part relative to a bottom surface of the drawer part; and a driving device located in the front panel door and configured to supply driving force to move the elevation device, wherein the elevation device comprises a rotation shaft disposed in a fixed position in the accommodation space of the drawer part in the front-rear direction, and wherein the driving device comprises: a motor assembly disposed in the front panel door to generate a driving force to drive the elevation device, a fixing shaft disposed at a position corresponding to the rotation shaft in the front panel door in the front-rear direction and rotatably connected to the motor assembly and the rotation shaft, and a connection member that is rotatably connected to the fixing shaft and the rotation shaft to transmit the driving force to the rotation shaft through the fixing shaft by a rotation force.

21. The refrigerator according to claim 20, wherein the elevation device comprises a plurality of rods disposed in the accommodation space in a left-right direction, the plurality of rods comprising a pair of first rods and a pair of second rods, wherein each of the pair of first rods and the pair of second rods are arranged in parallel and rotatably coupled to and cross each other, and wherein the rotation shaft is disposed between the pair of the first rods in the front-rear direction and is connected to one end of the first rod at the fixed position in the drawer part such that the rotation

shaft rotates with the pair of the first rods based on the fixing shaft rotating.

22. The refrigerator according to claim 21, wherein one of the pair of first rods adjacent to the front panel door comprises a rod protrusion that protrudes toward the front panel door in a same extension line with the rotation shaft, and wherein the connection member is coupled the fixing shaft to the rod protrusion.

23. The refrigerator according to claim 22, wherein the drawer part comprises a drawer opening on a front side surface, and wherein the rod protrusion is exposed through the drawer opening to be coupled to the connection member.
